

Functional characterisation of neuropeptides that act as ligands for both calcitonin-type and pigment-dispersing factor-type receptors in a deuterostome

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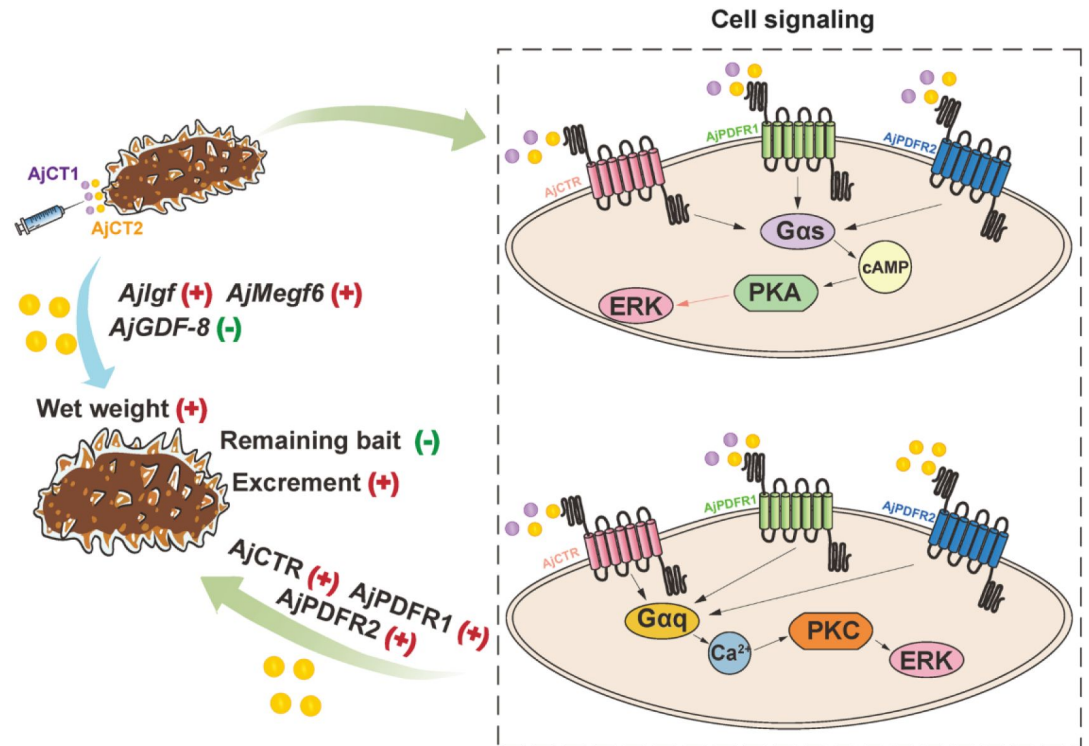
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Abstract

The calcitonin (CT) family of related peptides exert their diverse physiological effects in mammals via two G-protein coupled receptors, CTR and the CTR-like receptor CLR. Analysis of the phylogenetic distribution of CT-type signaling has revealed the occurrence of CT-type peptides and CTR/CLR-type proteins in deuterostome and protostome invertebrates. Furthermore, experimental studies have revealed that in the protostome *Drosophila melanogaster* the CT-like peptide DH₃₁ can act as a ligand for a CTR/CLR-type receptor and a pigment-dispersing factor (PDF) receptor. Here we investigated the signaling mechanisms and functions of CT-type neuropeptides in a deuterostome invertebrate, the sea cucumber *Apostichopus japonicus* (phylum Echinodermata). In *A. japonicus*, a single gene encodes two CT-type peptides (AjCT1 and AjCT2), both of which act as ligands for a CTR/CLR-type receptor (AjCTR) and two PDF-type receptors (AjPDFR1, AjPDFR2), but with differential activation of downstream cAMP/PKA, Gαq/Ca²⁺/PKC and ERK1/2 signaling pathways. Analysis of the expression of the gene encoding AjCT1 and AjCT2 revealed transcripts in a variety of organ systems, but with highest expression in the circumoral nervous system. *In vitro* pharmacological experiments revealed that AjCT1 and/or AjCT2 cause dose-dependent relaxation of longitudinal body wall muscle and intestine preparations. Furthermore, *in vivo* pharmacological experiments and loss-of-function tests revealed a potential physiological role for AjCT2 signaling in promoting feeding and growth in *A. japonicus*. This is the first study to obtain evidence that CT-type peptides can act as ligands for both CTR/CLR-type and PDF-type receptors in a deuterostome. Furthermore, because of the economic importance of *A. japonicus* as a foodstuff, discovery of the potential role for CT-type peptides as regulators of feeding and growth in this species may provide a basis for practical applications in aquaculture.

Graphical abstract



A schematic showing proposed molecular mechanisms by which CT-type neuropeptides regulate feeding and growth in *A. japonicus*. AjCT1 and AjCT2 are represented by purple and yellow circles, respectively. For cell signaling, black arrows indicate that this process is activated, whilst the pink arrow indicates that the ERK1/2 is activated in the AjCT1/AjPDR1, AjCT2/AjPDR1 and AjCT2/AjPDR2 signaling pathways.

eLife assessment

This **valuable** study characterises receptors for calcitonin-related peptides from a deuterostomian animal, the echinoderm *Apostichopus japonicus*, by a combination of heterologous expression, pharmacological experiments, and the quantification of gene-expression levels. The authors provide **solid** evidence for a functional calcitonin-related peptide system in the sea cucumber, but some of the phylogenetic and statistical analyses and the evidence of a physiological function are **incomplete**. This work should be of interest to scientists studying the signaling pathways, functions, and evolution of neuropeptides, and could be of relevance to improving the culture conditions of this economically key species.

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1. Introduction

The peptide calcitonin (CT) was first identified as a hormone that regulates blood calcium levels and subsequently a family of CT-related neuropeptides was discovered in mammals, including calcitonin gene related peptides (CGRPs), amylin (AMY), adrenomedullin (AM), and CT receptor-stimulating peptide (CRSP) (Copp et al., 1962; Sekiguchi et al., 2009 [↗](#); Sekiguchi et al., 2016 [↗](#)). Members of the CT-type family of peptides typically have a disulfide bond in their N-terminal region (Cai et al., 2018 [↗](#)) and exert their physiological effects in mammals by binding to two related G protein-coupled receptors: CTR (CT receptor) and CLR (CTR-like receptor) (Martins et al., 2014 [↗](#)). However, receptor activity-modifying proteins (RAMPs) acting as accessory proteins are key determinants of ligand selectivity for CTR and CLR (Mclatchie et al., 1998). Thus, formation of a complex with a RAMP is essential for CLR function, whereas RAMPs are not required for CTR ligand recognition (Kuwasako et al., 2011 [↗](#); Bomberger et al., 2012). Furthermore, binding of CT to CTR in the absence of RAMPs (Katafuchi et al., 2009 [↗](#)) can activate signaling via several downstream pathways, including cAMP accumulation, Ca^{2+} mobilization, and ERK activation (Chen et al., 1998 [↗](#); Pondel, 2000 [↗](#); Walker et al., 2010; Andreassen et al., 2014).

Transcriptome/genome sequencing has enabled analysis of the phylogenetic distribution of CT-type signaling systems, indicating that the evolutionary origin of CT-type peptides and CTR/CLR-related receptors can be traced back to the urbilaterian common ancestor of deuterostomes and protostomes (Mirabeau and Joly, 2013; Jekely, 2013). Furthermore, in protostomes two different types of CT-related peptides have been identified. Firstly, peptides that are structurally similar to vertebrate CT-type peptides in having an N-terminal disulphide bridge and secondly, diuretic hormone 31 (DH₃₁)-type peptides that lack an N-terminal disulphide bridge. Furthermore, phylogenetic analysis indicates that gene duplication in a common ancestor of the protostomes gave rise to genes encoding CT-like peptides with a disulphide bridge and DH₃₁-type peptides, which were then differentially retained in protostome taxa (Cai et al., 2018 [↗](#)). Receptors for CT-like peptides with a disulphide bridge have been characterised in a molluscan species, the oyster *Crassostrea gigas*, which has seven CTR/CLR-type receptors. Two of these receptors, named Cg-CTR and Cragi-CTR2, were shown to be specifically activated by the CT-type neuropeptides Cragi-CT1b and Cragi-CT2, respectively, thereby mediating intracellular Ca^{2+} mobilization and cAMP accumulation (Schwartz et al., 2019 [↗](#)). CT-like peptides with a disulphide bridge are not present in the insect *Drosophila melanogaster*. However, receptors that are activated by DH₃₁ have been identified in this species and, interestingly, DH₃₁ can act as a ligand for two types of receptors. Thus, DH₃₁ acts as a ligand for a CTR/CLR-related protein (CG17415), but it also acts as a ligand for a protein (CG13758) that is the receptor for pigment-dispersing factor (PDF) (Mertens et al., 2005 [↗](#); Shafer et al., 2008). However, the evolutionary and functional significance of the occurrence of dual receptor pathways for DH₃₁ in *Drosophila* are not known.

Currently, relatively little is known about CT-type signaling systems in deuterostome invertebrates. Analysis of transcriptome/genome sequence data has enabled identification of genes/transcripts encoding CT-type peptides and candidate CTR/CLR-type receptors for these peptides in several deuterostome invertebrate taxa, including urochordates (e.g. *Ciona intestinalis*), cephalochordates (e.g. *Branchiostoma floridae*), hemichordates (e.g. *Saccoglossus kowalevskii*) and echinoderms (e.g. *Strongylocentrotus purpuratus*) (Burke et al., 2006; Sekiguchi et al., 2009 [↗](#); Sekiguchi et al., 2016 [↗](#); Rowe and Elphick, 2012; Martins et al., 2014 [↗](#)). Furthermore, analysis of the expression and action of CT-type peptides identified in the starfish *Asterias rubens* and *Patiria pectinifera* has revealed that they act as muscle relaxants (Cai et al., 2018 [↗](#); Semmens et al., 2016). However, experimental confirmation of ligand-receptor partners has thus far only been reported for the cephalochordate *B. floridae*, where three CT-type peptides were shown to act as ligands for a CTR/CLR-type receptor when it is co-expressed with RAMP-type proteins (Sekiguchi et al., 2016 [↗](#)).

To gain further insights into the mechanisms of CT-type signaling in invertebrate deuterostomes, the aim of this study was to functionally characterise CT-type peptides and their receptors in an echinoderm – the sea cucumber *Apostichopus japonicus*.

- *A. japonicus* is an economically important species in Asian aquaculture (Pangestuti and Arifin, 2018 [↗](#); Guo et al., 2020 [↗](#)) and a species representative of the phylum Echinodermata. As a phylum of deuterostome invertebrates, echinoderms form a sister clade to the chordates together with hemichordates and xenoturbellids (Bromham and Degnan, 1999; Bourlat et al., 2006; Dunn et al., 2008) and as such they occupy a key “intermediate” phylogenetic position relative to the protostome invertebrates and vertebrates (Zandawala et al., 2017), which make them attractive as experimental systems to address evolutionary questions. In our previous studies, we have established this species as a model system to obtain insights into the evolutionary history and comparative physiology of neuropeptide signaling systems in bilaterians (Chen et al., 2019 [↗](#); Li et al., 2022 [↗](#); Zheng et al., 2022 [↗](#); Cong et al., 2023 [↗](#); Zheng et al., 2024 [↗](#)). A gene encoding two CT-type peptides (AjCT1 and AjCT2) has been identified in *A. japonicus* (Chen et al., 2019 [↗](#)) and a candidate receptor (AjCTR) for these peptides has also been predicted (Huang et al., 2021 [↗](#)). Furthermore, relevant to the discovery that the CT-like peptide DH₃₁ can act as a ligand for both a CTR/CLR-type receptor and a PDF-type receptor in *Drosophila*, the sequences of two PDF-type receptors have been identified in *A. japonicus* (AjPDFR1 and AjPDFR2) (Huang et al., 2021 [↗](#)).

The first aim of this study was to investigate if AjCT1 and/or AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2 and to characterise the downstream signaling pathways that are activated when AjCT1 and/or AjCT2 bind to their receptor(s). The second aim was to investigate the physiological roles of AjCT1 and AjCT2 in *A. japonicus* by analysing the expression of the gene encoding these peptides, examining the *in vitro* and *in vivo* pharmacological effects of these peptides and performing loss-of-function tests. The functional characterisation of CT-type signaling in *A. japonicus* reported in this study provides new insights into the evolution and comparative physiology of neuropeptides and a basis for application of neuropeptide-based strategies to improve aquaculture methods for this economically important species.

2. Results

2.1 Characterization and phylogenetic analysis of AjCT1/2

The complete cDNA sequences of the CT-type neuropeptide precursors *AjCTP1* and *AjCTP2* were confirmed. The coding sequence of *AjCTP2* is shorter than *AjCTP1* (Figure 1A [↗](#)) but both transcripts encode precursor proteins that comprise the same predicted 23-residue N-terminal signal peptide (Figure 1-figure supplement 1A [↗](#)). *AjCTP1* encodes both putative CT-type neuropeptides (AjCT1 and AjCT2), while *AjCTP2* only encodes AjCT2 (Figure 1A [↗](#), Figure 1-figure supplement 1A [↗](#) and 1B [↗](#)). Exon and intron structure analysis showed that *AjCTP1* comprises four exons, with AjCT1 and AjCT2 encoded in the third and fourth exons, respectively. *AjCTP2* comprises three exons, excluding the third exon that encodes AjCT1 (Figure 1A [↗](#)). Phylogenetic analysis revealed that *AjCTP1* and *AjCTP2* are closely related to CT-type precursors from other sea cucumbers (*Holothuria glaberrima* and *Holothuria scabra*), and are positioned within a clade that also contains CT-type precursors from other echinoderms (sea urchin *S. purpuratus*, starfish *A. rubens*). Furthermore, with vasopressin/oxytocin-type neuropeptide precursors included as an outgroup, the echinoderm CT-type precursors are positioned within a clade that also comprises CT-type precursors from chordates and DH₃₁/CT-type precursors from protostomes (Figure 1B [↗](#)). The predicted tertiary structures of the AjCTP1 and AjCTP2 proteins are shown in Figure 1-figure supplement 1C [↗](#). Sequence alignment of CT-related neuropeptides revealed that, with the exception of DH₃₁-type peptides, the N-terminal region of CT-related peptides has a pair of

conserved cysteine residues that are known or predicted to form a disulfide bond. Another conserved feature is a C-terminal proline residue, with the exception of CGRPs (**Figure 1-figure supplement 2**). Accordingly, both AjCT1 and AjCT2 have these two conserved characteristics described above.

2.2 Characterization and phylogenetic analysis of three candidate receptors for AjCT1/2

The sequences of three candidate receptors for AjCT1 and AjCT2 were obtained by RACE and phosphorylation sites were predicted, as shown in **Figure 1-figure supplement 3A**, **3B**, and **3C**. Phylogenetic analysis revealed that CTR/CLR-type receptors divided into two clades and AjCTR was positioned in a clade with other deuterostome CTR/CLR-type receptors, whilst the other clade comprised CTR/CLR-type receptors from protostomes. Similarly, AjPDFR1 and AjPDFR2 were positioned in a clade comprising PDF-type receptors from other deuterostomes, whilst another clade comprised PDF-type receptors from protostomes (**Figure 1C**). Domain prediction of the candidate receptors showed that AjCTR had HRM (Hormone receptor) and 7tm_GPCRs (seven-transmembrane G protein-coupled receptor) superfamily domains (**Figure 1-figure supplement 4A**), whereas AjPDFR1 and AjPDFR2 only presented the 7tm_GPCRs superfamily domain (**Figure 1-figure supplement 4B**). To further investigate an orthologous relationship of the candidate receptors with homologous receptors in other echinoderms, we conducted a gene synteny analysis in *A. japonicus*, *A. rubens* and *L. variegatus*. The neighboring genes of *AjCTR* included *GOLGA4*, *NeuroD*, *CRF2*, *ppGalNAc-T*, *URB3*, *SLC40A1*, *CMAH* and *MAGP*. The relative positions of these genes were different from those in *A. rubens* and *L. variegatus* (**Figure 1-figure supplement 4C**). The same situation was also observed for the synteny analysis of PDFRs.

2.3 AjCT1 and AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2

To investigate if AjCT1 and/or AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2, here, two receptor assay methods were employed. As shown in **Figure 2A**, both AjCT1 and AjCT2 (10^{-6} M) caused a significant increase in CRE luciferase activity in HEK293T cells transfected with AjCTR, AjPDFR1 or AjPDFR2 (**Table 1**). Furthermore, this effect was inhibited by pre-incubation of cells with the PKA inhibitor H89. The intracellular accumulation of cAMP was also measured, and this showed that AjCT1 and AjCT2 (10^{-9} – 10^{-5} M) caused a dose-dependent increase in cAMP in HEK293T cells transfected with AjCTR, AjPDFR1 or AjPDFR2 (**Figure 2B**). However, when exposed to AjCT1 or AjCT2, the maximum accumulation of intracellular cAMP was higher in cells transfected with *AjCTR* than in cells transfected with *AjPDFR1* and *AjPDFR2* (**Figure 2B**). Measurement of SRE luciferase activity revealed that AjCT1 and AjCT2 (10^{-6} M) administration evoked an increase in activity in HEK293T cells transfected with *AjCTR* and *AjPDFR1* (**Figure 3A**). However, in HEK293T cells transfected with *AjPDFR2*, an increase in SRE luciferase activity was only observed with AjCT2, not with AjCT1 (**Table 1**). Furthermore, as shown in **Figure 3A**, AjCT1- and AjCT2-induced increases in SRE luciferase activity were not observed in cells that had been pre-incubated with FR900359 (Gaq protein inhibitor) and Gö 6983 (PKC inhibitor). As shown in **Figure 3-figure supplement 1**, receptors localized in the cell membrane, but after exposure to AjCT1 and AjCT2 (10^{-6} M) for 15 min, internalization of pEGFP-N1/*AjCTR*, pEGFP-N1/*AjPDFR1* and pEGFP-N1/*AjPDFR2* was observed in transfected HEK293T cells (**Figure 3B**). Taken together, all these findings indicate that AjCT1 and AjCT2 can act as ligands for the three candidate receptors and then activate cAMP/PKA signaling and Gaq/ Ca^{2+} /PKC signaling in all combinations, with the exception of AjCT1/*AjPDFR2* which only activates the cAMP/PKA cascade.

To investigate whether the activated signaling pathways may exert downstream effects via the ERK–MAPK pathway, the phosphorylation levels of ERK1/2 were analysed in transfected HEK293T cells. pERK1/2 levels in cells transfected with *AjCTR* or *AjPDFR1*, but not *AjPDFR2*, were significantly increased after a 5 min incubation with AjCT1 (**Figure 4A**). Furthermore, this effect was blocked by the PKC inhibitor in AjCTR-expressing cells, and this effect was blocked by both

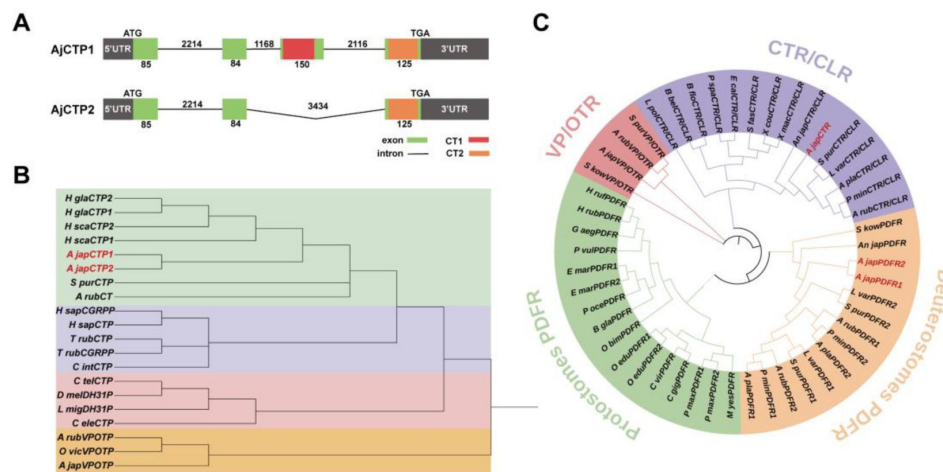
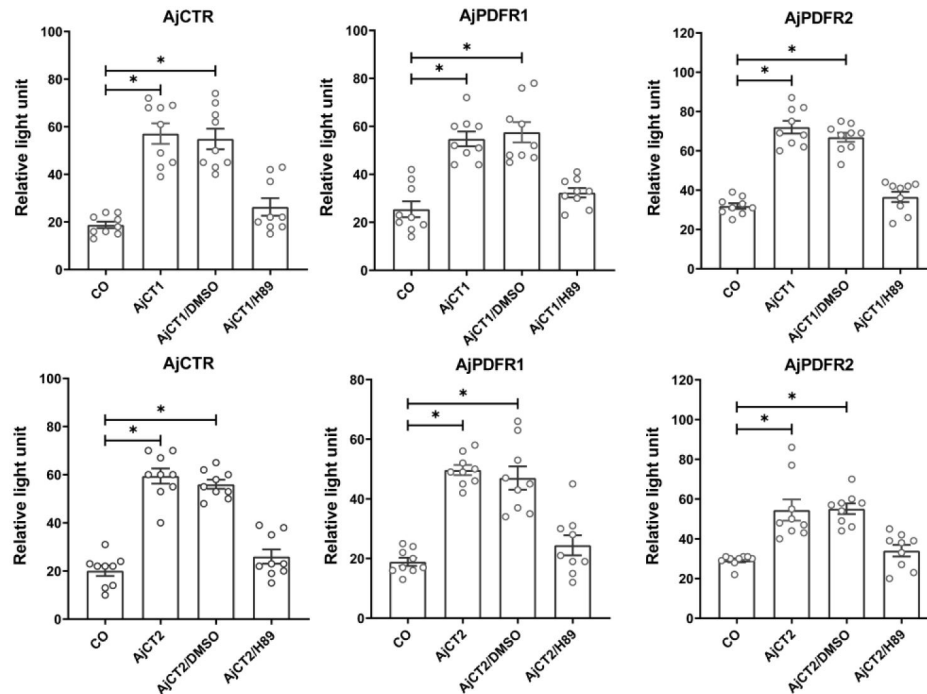


Figure 1.

CT-type neuropeptide precursors and the candidate receptors in *A. japonicus*. (A) Alternative splicing schematic of *AjCTP1* and *AjCTP2*. Exons are represented by green rectangles, introns are represented by lines, *AjCT1* and *AjCT2* are represented by red and orange rectangles respectively, upstream and downstream non-coding regions are represented by gray rectangles. The numbers represent the length of exons or introns. (B) Phylogenetic analysis of *AjCTP1* and *AjCTP2* and CT-related neuropeptide precursors from other bilaterians. CT-type precursors from deuterostomes are shown in green (echinoderms) and purple (chordates), CT-type precursors from protostomes are shown in pink and VP/OT-type neuropeptide precursors (outgroup) are shown in orange. (C) Phylogenetic analysis of relationships of *AjCTR*, *AjPDFR1* and *AjPDFR2* (red letters) with CT-type and PDF-type receptors in other taxa (black letters), using VP/OT-type receptors as an outgroup. CTR/CLR-type receptors are shown in purple, deuterostome PDFRs are shown in yellow, protostome PDFRs are shown in green and VP/OTRs are shown in pink. Full species names and accession numbers are listed in Supplementary Table 3.

A



B

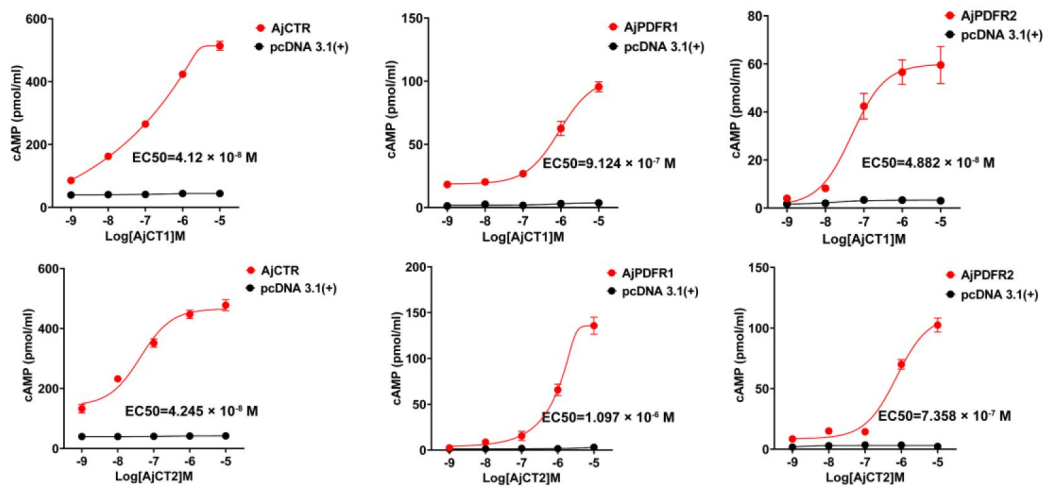


Figure 2.

Pharmacological characterisation of AjCT1 and AjCT2 as ligands for the *A. japonicus* receptors AjCTR, AjPDFR1 and AjPDFR2. (A) CRE-driven luciferase activity measured in HEK293T cells transfected with one of the three candidate receptors after exposure to AjCT1 or AjCT2 or neuropeptide + DMSO or neuropeptide + H89 (10 μ M). The neuropeptide concentration used here was 10⁻⁶ M. (B) Measurement of cAMP accumulation after exposure to AjCT1 or AjCT2 (10⁻⁹ – 10⁻⁵ M). No cAMP elevation is observed in control experiments where HEK293T cells were transfected with empty pcDNA 3.1(+) (black circle). The EC₅₀ values are shown in the figure. Error bars represent SEM for three independent experiments. * indicates a statistically significant difference with $p < 0.05$.

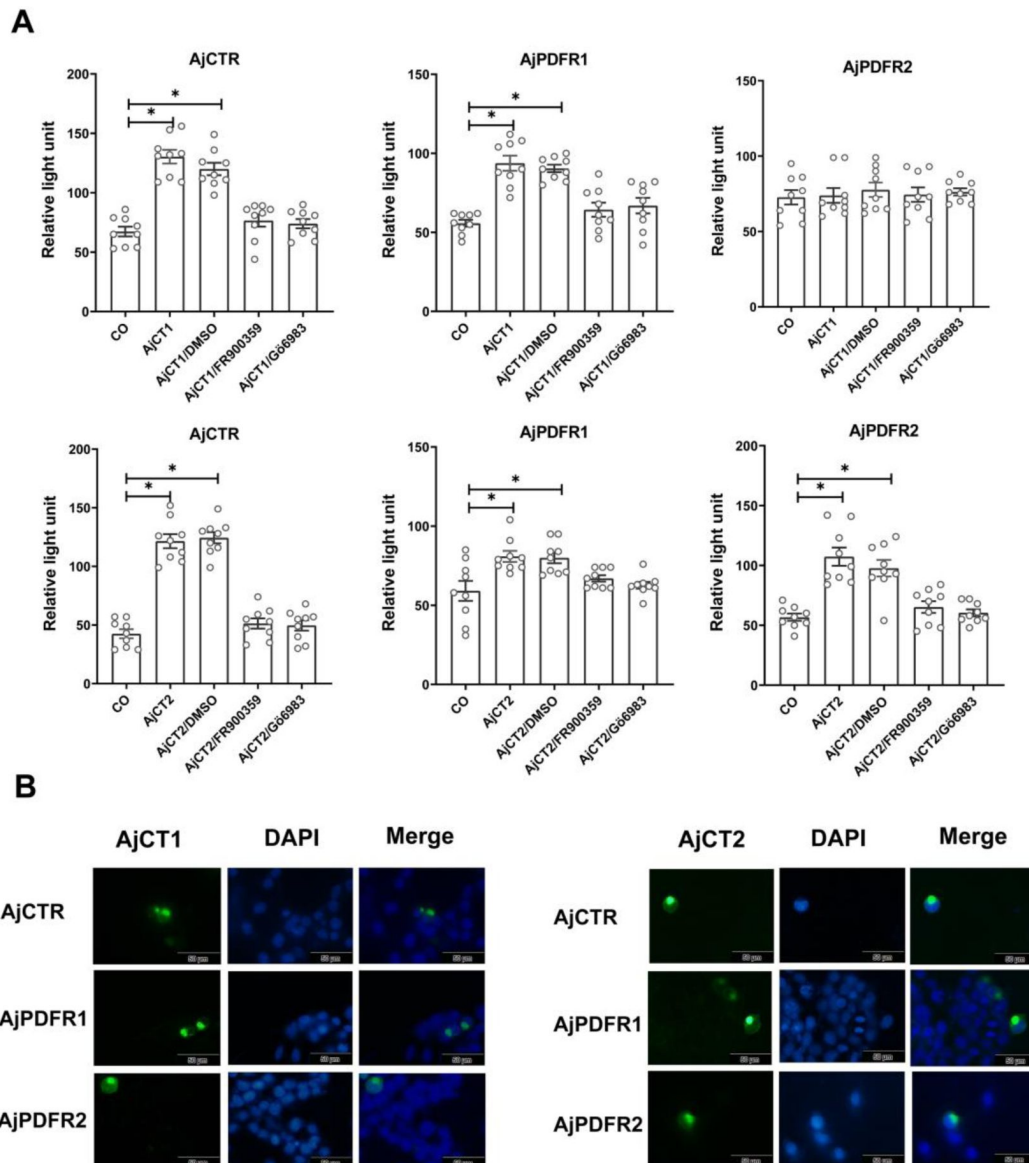


Figure 3.

SRE-driven luciferase activity and receptor internalization in cells transfected with AjCTR, AjPDFR1 or AjPDFR2 and exposed to AjCT1 or AjCT2. (A) SRE-driven luciferase activity measured after incubation with AjCT1, AjCT2, neuropeptide (10^{-6} M)/DMSO, neuropeptide (10^{-6} M)/FR900359 (1 μ M) and neuropeptide (10^{-6} M)/Gö 6983 (1 μ M). (B) Localization and internalization of AjCTR, AjPDFR1 and AjPDFR2 after 15 min treatment with 10^{-6} M AjCT1 or AjCT2. Green fluorescence represents the localization of pEGFP-N1/receptors in cells. The cell nucleus probe (DAPI) was used for cell nucleus staining. Scale bar: 50 μ m. Error bars represent SEM for three independent experiments. * indicates a statistically significant difference with $p < 0.05$. ns, indicates no significant difference. All pictures are representative of at least three independent experiments.

Assays Receptors	cAMP/PKA		ERK1/2		Gαq/Ca ²⁺ /PKC		ERK1/2	
	AjCT1	AjCT2	AjCT1	AjCT2	AjCT1	AjCT2	AjCT1	AjCT2
<i>AjCTR</i>	√	√	×	×	√	√	√	√
<i>AjPDR1</i>	√	√	√	√	√	√	√	√
<i>AjPDR2</i>	√	√	×	√	×	√	×	√

A tick indicates activation and the cross indicates no activation.

Table 1.

Activation of signaling pathways

PKA and PKC inhibitors in AjPDFR1-expressing cells (**Figure 4-figure supplement 1A** [↗](#)). pERK1/2 levels in cells transfected with *AjCTR* or *AjPDFR1* or *AjPDFR2*, were significantly increased after a 5 min incubation with AjCT2 (**Figure 4B** [↗](#)). Moreover, this effect persisted until 15 min after exposure to AjCT in both *AjCTR*-transfected cells and *AjPDFR1*-transfected cells but not in *AjPDFR2*-transfected cells (**Figure 4B** [↗](#), **Table 1** [↗](#)). Furthermore, these effects were blocked by PKA and PKC inhibitors in AjPDFR1 and AjPDFR2 expressing cells, whereas in AjCTR expressing cells, peptide-induced pERK1/2 activity was only blocked by a PKC inhibitor (**Figure 4-figure supplement 1B** [↗](#)). These results indicate that, with the exception of AjCT1/AjPDFR2, both AjCT1 and AjCT2 can activate the ERK1/2 cascade by binding to the candidate receptors.

2.4 Localization of AjCTP1/2 expression in *A. japonicus*

qRT-PCR analysis revealed that *AjCTP1/2* had the highest expression level in the circumoral nervous system (CNS) compared with other tissues ($p < 0.05$) (**Figure 5A** [↗](#)).

Informed by these findings, mRNA *in situ* hybridization was used to visualize transcripts in the CNS and the *AjCTP1/2* positive signals were observed (**Figure 5B** [↗](#)). No positive staining was detected in the CNS incubated with the sense probe (**Figure 5-figure supplement 1** [↗](#)).

2.5 AjCT1 and AjCT2 causes dose-dependent relaxation of longitudinal muscle and intestine preparations from *A. japonicus*

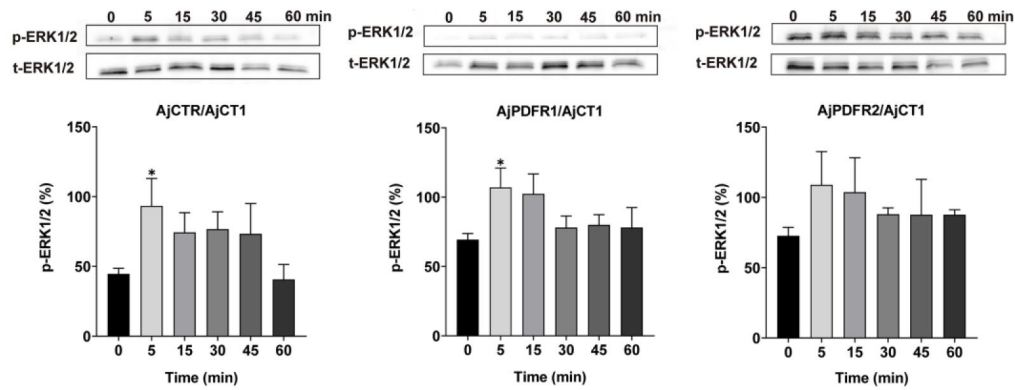
Both AjCT1 and AjCT2 caused dose-dependent relaxation of longitudinal muscles from *A. japonicus* at 10^{-10} – 10^{-6} M. At the highest concentration tested (10^{-6} M), AjCT1 and AjCT2 caused $30.07 \pm 7.49\%$ and $32.50 \pm 14.49\%$ reversal of ACh-induced contraction, respectively (**Figure 6A, 6B** [↗](#)). AjCT1 had no effect on the contractile state of intestine preparations, even at concentrations as high as 10^{-5} M (**Figure 6C** [↗](#)). Unlike AjCT1, AjCT2 caused dose-dependent relaxation of intestine preparations, with $26.33 \pm 8.75\%$ reversal of ACh-induced contraction observed at 10^{-5} M (**Figure 6D** [↗](#)).

2.6 AjCT2 affects feeding and growth in *A. japonicus*

To investigate the effects of AjCT1 and AjCT2 *in vivo*, a long-term injection experiment was carried out. As shown in **Figure 7A** [↗](#), there were no significant changes in wet mass in the CT1L, CT1H and CT2L groups ($p \geq 0.05$), but in the CT2H group there was a significant increase in wet weight after 24 days of injection in comparison with the control group (CO) ($p < 0.05$). Correspondingly, the mass of faeces increased and the residual bait mass decreased (**Figure 7A-figure supplement 1** [↗](#)). Furthermore, the relative intestinal expression levels of several growth factors were investigated and the results showed that *AjGDF-8*, a growth inhibitory factor, was significantly down-regulated in all the experimental groups. In contrast, the relative expression levels of *AjIgf* and *AjMegf6*, growth promoting factors, were significantly up-regulated in intestines (**Figure 7B** [↗](#)).

Subsequently, the relative intestinal expression levels of AjCT1 and AjCT2 receptors were further tested for the CT2H group. The results showed that the relative expression levels of all three receptors were significantly up-regulated, with the ranked fold change being *AjCTR* > *AjPDFR1* > *AjPDFR2* (**Figure 7C** [↗](#)). Conversely, the relative expression levels of the three receptor genes were significantly down-regulated in si*AjCTP1-1* treated animals (**Figure 8A, 8B** [↗](#) and **8C** [↗](#)). The interference rate for si*AjCTP1/2-1* was 66 % (**Figure 8-figure supplement 1** [↗](#)).

A



B

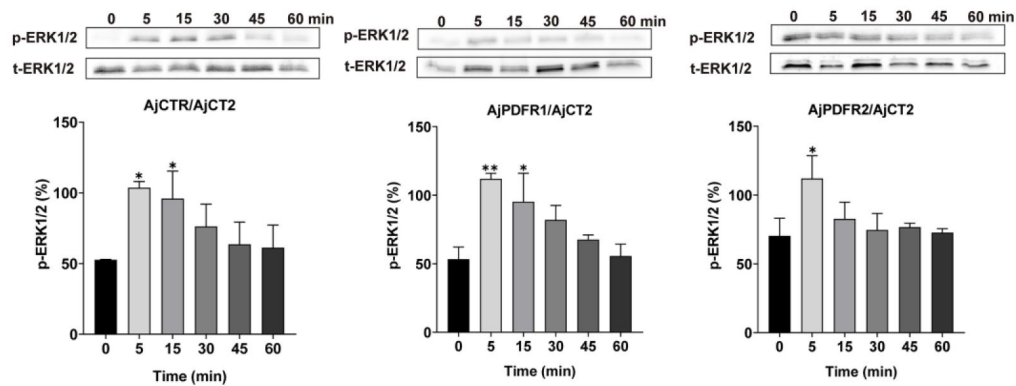


Figure 4.

ERK1/2 activity in cells transfected with AjCTR, AjPDFR1 or AjPDFR2 and exposed to AjCT1 or AjCT2. (A) and (B) show the immunoblot intensity of representative phosphorylation bands. The transfected HEK293T cells were incubated with AjCT1 (10^{-6} M) or AjCT2 (10^{-6} M) for 0, 5, 15, 30, 45 and 60 min. The p-ERK1/2 was normalized based on t-ERK1/2. Error bars represent SEM for three independent experiments. Statistically significant differences are indicated as follows: *: $p < 0.05$, **: $p < 0.01$.

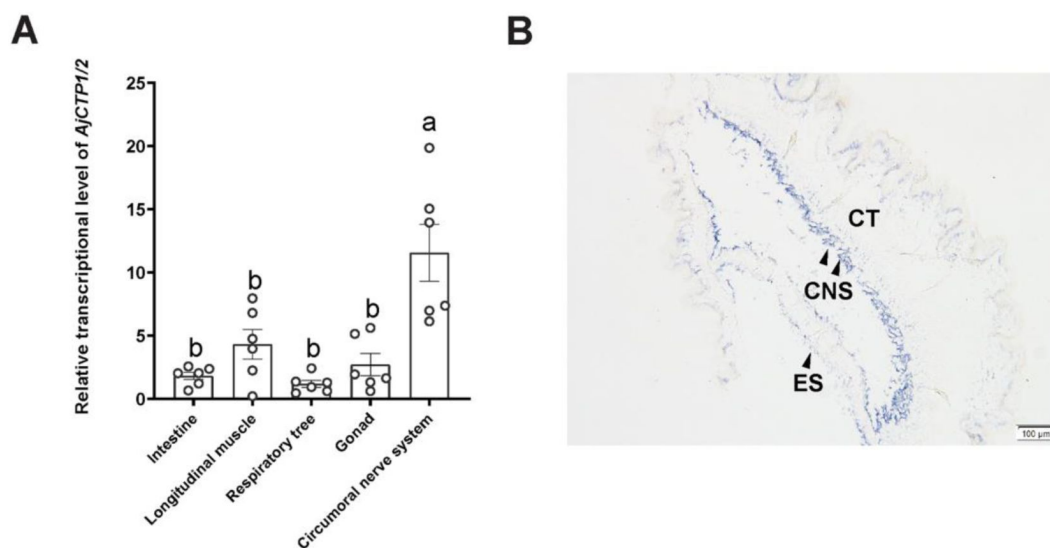


Figure 5.

The expression profile of *AjCTP1/2* in *A. japonicus*. (A) Analysis of the transcript expression level of *AjCTP1/2* in different tissues. Lowercase letters a and b above columns indicate the statistical difference at $p < 0.05$; the expression level of *AjCTP1/2* in CNS was significantly higher than that in other tissues. Mean values with standard deviations ($n = 5$) are shown. (B) Localization of *AjCTP1/2* in CNS using mRNA *in situ* hybridization. CNS, circumoral nervous system; CT, connective tissue; ES, epidermis. Scale bar: 100 μ m.

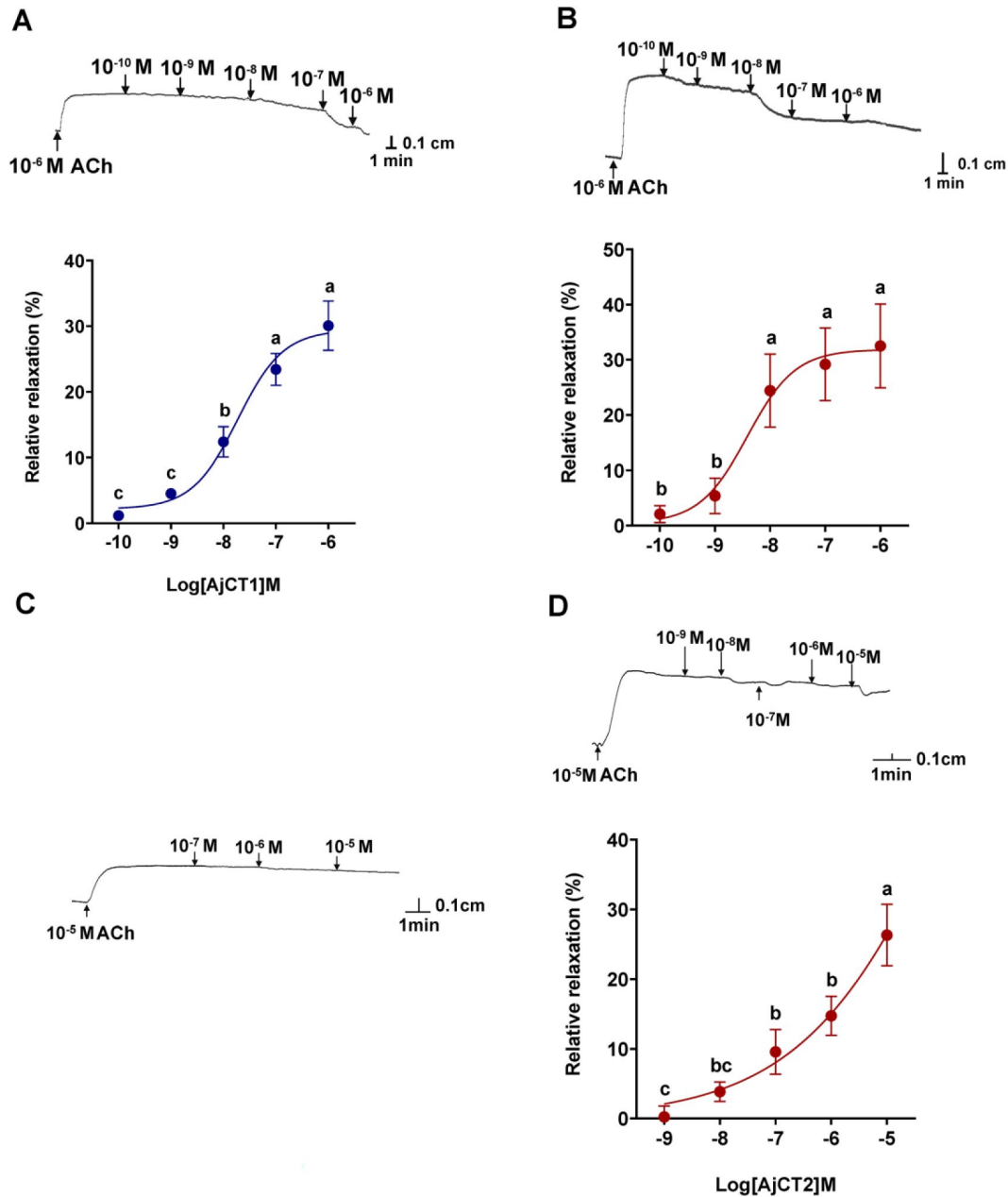


Figure 6.

Pharmacological effects of AjCT1 and AjCT2 on longitudinal muscle and intestine preparations from *A. japonicus*. (A) and (B) show representative recordings of the relaxing effects of AjCT1 and AjCT2 on longitudinal muscle preparations. The graphs show the dose-dependent relaxing effects of AjCT1 and AjCT2 (10^{-10} – 10^{-6} M), calculated as the percentage reversal of the contracting effect of 10^{-6} M ACh. (C) and (D) show representative recordings of experiments in which AjCT1 (10^{-7} – 10^{-5} M) and AjCT2 (10^{-9} – 10^{-5} M) were tested on intestine preparations. AjCT1 had no effect, but AjCT2 caused relaxation and the graph shows the dose-dependent relaxing effect of AjCT2 (10^{-9} – 10^{-5} M), calculated as the percentage reversal of the contracting effect of 10^{-5} M ACh. Mean values $\pm$ SEM were determined from three preparations. Lowercase letters a, b and c above columns indicate the statistical difference at $p < 0.05$, whilst there is no significant difference between bc and c.

Figure 7.

In vivo pharmacological tests revealed that AjCT1 and AjCT2 affect feeding and growth-related gene expression in *A. japonicus*. (A) Body mass in different groups before and after injection of AjCT1, AjCT2 or sterilized seawater. (B) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in different groups after injection of AjCT1 or AjCT2 or sterilized seawater (CO) in *A. japonicus*. (C) The relative intestinal expression levels of AjCT1/2 receptors (*AjCTR*, *AjPDRF1* and *AjPDRF2*) in CT2H and CO groups after 24 days injection experiment. Values were mean $\pm$ SEM (n = 5). * indicates statistically significant differences with $p < 0.05$, ** indicates statistically significant differences with $p < 0.01$. CO: control group (sterilized seawater); CT1L/CT2L: AjCT1/AjCT2 low concentration group (5×10^{-3} mg/mL); CT1H/CT2H: AjCT1/AjCT2 high concentration group (5×10^{-1} mg/mL).

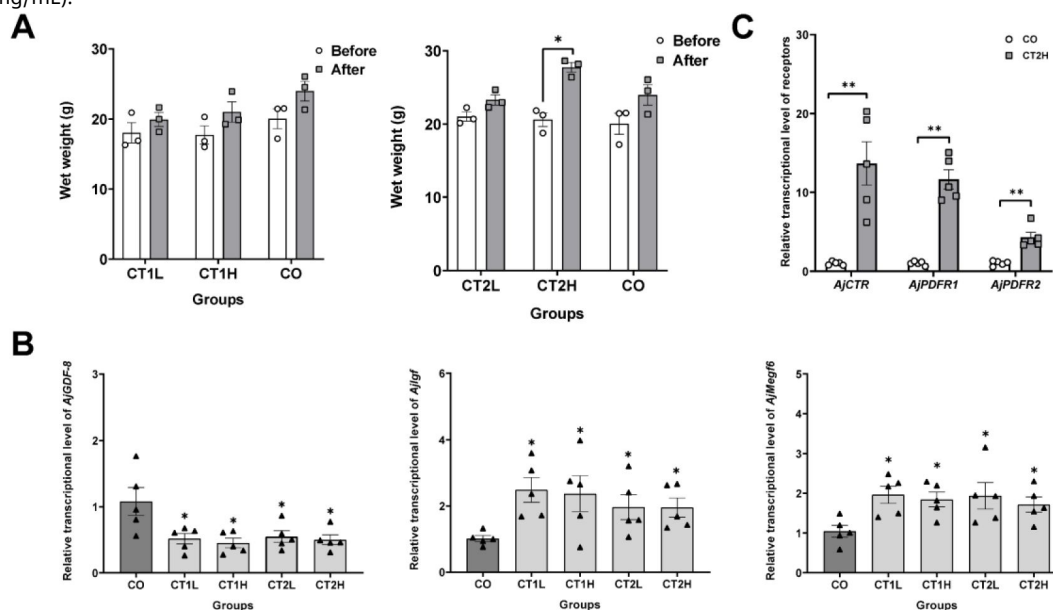
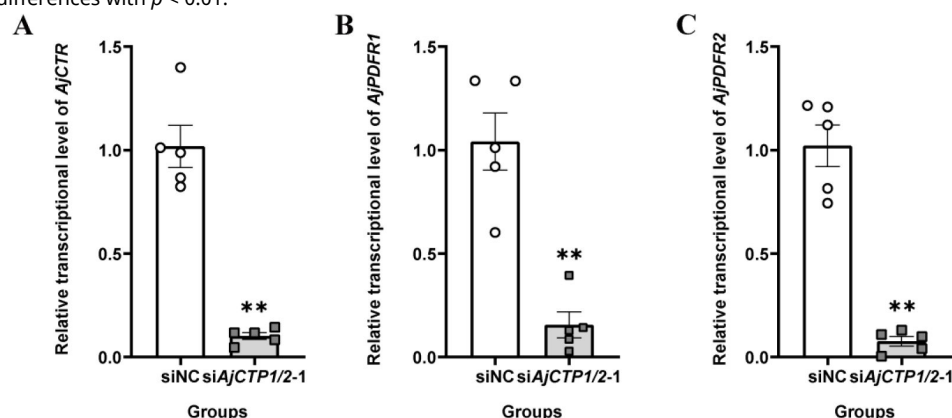


Figure 8.

The relative intestinal expression levels of AjCT1/2 receptors after *AjCTP1/2-1* knockdown. (A), (B) and (C) show the relative expression levels of *AjCTR*, *AjPDRF1* and *AjPDRF2* respectively. Values are means $\pm$ SEM (n = 5). ** indicates statistically significant differences with $p < 0.01$.



3. Discussion

Previous studies have revealed that the CT-type neuropeptide DH₃₁ can act as a ligand for a CTR/CLR-type receptor and a pigment-dispersing factor (PDF) receptor in a protostome – the insect *D. melanogaster* (Mertens et al., 2005 [↗](#); Shafer et al., 2008; Schwartz et al., 2019 [↗](#)). Here we show that two CT-type peptides (AjCT1 and AjCT2) act as ligands for a CTR/CLR-type receptor and two pigment-dispersing factor (PDF) type receptors in a deuterostome – the sea cucumber *A. japonicus* (phylum Echinodermata). This is an important finding because it indicates that the ability of CT-type neuropeptides to act as ligands for both CTR/CLR-type receptors and PDF-type receptors is an evolutionarily ancient property that originated in a common ancestor of the Bilateria. However, further studies on a wider variety of both protostome (e.g. molluscs, annelids) and deuterostome taxa (e.g. other echinoderms, hemichordates) will be needed to establish if this is an evolutionarily conserved property of CT-type neuropeptides. Here, we also investigated the signal transduction pathways that are activated by AjCT1 and AjCT2 when these peptides bind to their receptors and then proceeded to investigate the expression and actions of these peptides to gain insights into their physiological/behavioral roles in *A. japonicus*, as discussed below.

3.1 Alternative RNA splicing gives rise to transcripts encoding one (AjCT2) or two (AjCT1, AjCT2) CT-type neuropeptides in *A. japonicus*

In vertebrates, alternative RNA splicing gives rise to transcripts encoding CT or CGRP (Amara et al., 1982; Morris et al., 1984). Analysis of transcriptome/genome sequence data from *A. japonicus* has revealed that in this species alternative RNA splicing gives rise to transcripts encoding one (AjCT2) or two (AjCT1, AjCT2) CT-type neuropeptides (Rowe et al., 2014; Zandawala et al., 2017; Chen et al., 2019 [↗](#); this study). A similar pattern of alternative RNA splicing has been reported for CT-type precursors in other sea cucumbers (*H. scabra* and *H. glaberrima*; class Holothuroidea) and in brittle stars (*Amphiura filiformis* and *Ophiopsila Aranea*; class Ophiuroidea) (Zandawala et al., 2017; Suwansa-Ard et al., 2018 [↗](#)). In contrast, CT-type precursors identified in other echinoderm classes (Asteroidea, Echinoidea and Crinoidea) comprise a single CT-type neuropeptide (Rowe and Elphick, 2012; Mirabeau and Joly, 2013; Rowe et al., 2014; Semmens et al., 2016; Cai et al., 2018 [↗](#); Aleotti et al., 2022 [↗](#)). However, it remains to be determined if this feature of CT-type precursors in sea cucumbers and brittle stars evolved independently or originated in a common ancestor of eleutherozoan echinoderms but with subsequent loss in Asteroidea and Echinoidea (Figure 9 [↗](#)).

Analysis of the sequences of AjCT1 and AjCT2 revealed the presence of a two cysteine residues in their N-terminal regions, which is an evolutionarily conserved feature of CT-type peptides. Furthermore, structural analysis of CT-type peptides in vertebrates and in starfish has revealed that these cysteines form a disulphide bridge in the mature peptides (Cai et al., 2018 [↗](#)) and therefore it is inferred that this post-translational modification also occurs in AjCT1 and AjCT2. Furthermore, the C-terminal residue of the AjCT1 and AjCT2 is predicted to be an amidated proline, which is a conserved post-translational modification of CT/DH₃₁-type peptides in many taxa, but not vertebrate CGRPs (Cai et al., 2018 [↗](#)).

3.2 Identification and characterisation of receptors for CT-type neuropeptides in *A. japonicus*

Informed by previous genome-wide analysis of GPCRs in *A. japonicus* (Huang et al., 2021 [↗](#)), we identified three candidate receptors for AjCT1 and/or AjCT2. Firstly, AjCTR, which is a homolog of CT/DH₃₁-type receptors that have been identified in other taxa, including vertebrates and insects (Garelja et al., 2022; Johnson et al., 2005; Toribio et al., 2003). Secondly, AjPDFR1 and AjPDFR2, which are homologs of the PDF-type receptor that has been shown to be activated by DH₃₁ in *D. melanogaster* (Mertens et al., 2005 [↗](#); Zandawala, 2012 [↗](#)).

Bilateria		Representative Species	CT-related		Receptors for CT		RAMP	Feeding /growth regulation
			precursor	predicted peptides	CTR/CLR	PDFR		
Deuterostomia	Vertebrata	<i>H. sapiens</i>		☆				I
	Urochordata	<i>C. intestinalis</i>						
	Cephalochordata	<i>B. floridae</i>						
	Hemichordata	<i>S. kowalevskii</i>		☆				
	Echinodermata	<i>A. japonicus</i>		★				P
		<i>S. purpuratus</i>						
		<i>A. rubens</i>						
		<i>A. filiformis</i>		★				
		<i>A. japonica</i>						
	Protostomia	Annelida	<i>C. telata</i>					
Mollusca		<i>C. gigas</i>						N
Arthropoda		<i>D. melanogaster</i>						
Nematoda		<i>C. elegans</i>						

Figure 9.

Summary of CT-related neuropeptides and their corresponding precursors, receptors, RAMPs and feeding/growth regulation in representative species. Echinoderms are boxed in purple dotted lines. The summary of *A. japonicus* is framed in red dotted lines. A green rectangle indicates the presence of neuropeptides or precursors. The red pentagram indicates that neuropeptides are generated by alternative splicing of precursor transcripts, unfilled pentagram represents a single gene encoding one neuropeptide, the filled pentagram represents a gene encoding two neuropeptides in tandem. For the receptors for CT, pink filled rectangles show that the presence of the corresponding receptors has been reported and unfilled rectangles indicate that their receptors have not been reported. Blue filled rectangles indicate that RAMPs have been identified in this species, blue rectangles with a line show that RAMPs have not been identified in this species. Yellow filled rectangles indicate that feeding/growth regulation by CT-type neuropeptides has been investigated and unfilled rectangles showed that there was no relevant study in this species. I: Inhibition; P: Promotion; N: Ineffective.

In mammals, CTR and CLR mediate effects of all CT-related neuropeptides, but their ligand specificity relies on the presence and types of receptor activity-modifying proteins (RAMPs) (McLatchie et al., 1998; Purdue et al., 2002; Conner et al., 2004; Hay et al., 2018). Thus, CTR acts as a functional receptor for CT-type peptides without an associated RAMP (Katafuchi et al., 2009), whereas activation of downstream signaling via CLR requires an associated RAMP (Hay et al., 2018). Here, we investigated the occurrence of RAMPs in *A. japonicus* using the tblastn method employed by Sekiguchi et al., (2016), which enabled identification of RAMPs in the cephalochordate *B. floridae*. However, no RAMPs were identified in *A. japonicus*, suggesting that AjCTR may be functionally more similar to CTR than CLR. Accordingly, tblastn analysis of other available echinoderm genomes (*A. rubens*, *S. purpuratus* and *A. japonica*), as well as transcriptome databases for brittle stars (*O. victoriae*, *O. aranea* and *A. filiformis*; no published genomes), also failed to identify RAMPs. Likewise, RAMPs have also not been found in protostomes (Schwartz et al., 2019). Therefore, RAMPs may be unique to chordates, although interestingly RAMPs have also not been identified in the urochordate *C. intestinalis* (Sekiguchi et al., 2009).

As highlighted above, here we show that the CT-type peptides AjCT1 and AjCT2 both act as ligands for a CTR/CLR-type receptor (AjCTR) and two PDF-type receptors (AjPDFR1, AjPDFR2) in *A. japonicus*. Furthermore, characterization of downstream signaling mechanisms revealed that both AjCT1 and AjCT2 trigger AjCTR-mediated cAMP accumulation, Ca²⁺ mobilization, and ERK signaling. These findings are consistent with previous studies on CT-type receptors in vertebrates and in the mollusc *Crassostrea gigas* (Pondel, 2000; Walker et al., 2010; Andreassen et al., 2014; Schwartz et al., 2019). Both AjCT1 and AjCT2 trigger cAMP accumulation, Ca²⁺ mobilization, and ERK signaling mediated by AjPDFR1. Both AjCT1 and AjCT2 triggered cAMP accumulation mediated by AjPDF2, but only AjCT2 (not AjCT1) triggered Ca²⁺ mobilization and ERK signaling mediated by AjPDFR2. These differences in the signal transduction mechanisms activated by AjCT1 and AjCT2 may account for differences in the pharmacological activities and physiological roles of these peptides, as discussed below.

3.3 AjCT1 and AjCT2 act as muscle relaxants in *A. japonicus*

Analysis of the expression of the gene encoding the precursors comprising AjCT2 or AjCT1 and AjCT2 revealed the occurrence of transcripts in the circumoral nervous system as well as in peripheral tissues/organs, including the longitudinal body wall muscle, digestive system and gonads. Accordingly, analysis of CT-type neuropeptide expression in other echinoderms has revealed a widespread pattern of expression in larval and adult starfish (*A. rubens*) and in juvenile and adult featherstars (*Antedon mediterranea*) (Mayorova et al., 2016; Cai et al., 2018; Aleotti et al., 2022). Several CT-related peptides act as muscle relaxants in vertebrates (Steiner et al., 1991; Pinto et al., 1996; Yoshimoto et al., 1998; Edvinsson et al., 2001; Golpon et al., 2001; Kandilci et al., 2008) and, accordingly, previous studies have revealed that CT-type neuropeptides cause relaxation of muscle preparations in starfish, including locomotory organs (tube feet) and the apical muscle, which is functionally similar to the longitudinal body wall muscle in sea cucumbers (Cai et al., 2018). Therefore, it was of interest to investigate if AjCT1 and/or AjCT2 act as muscle relaxants in *A. japonicus*. Both AjCT1 and AjCT2 caused dose-dependent relaxation of longitudinal muscle preparations, consistent with the relaxing effect of CT-type peptides on starfish apical muscle preparations. Furthermore, AjCT2, but not AjCT1, caused dose-dependent relaxation of intestine preparations from *A. japonicus*. Interestingly, the CT-type peptide ArCT does not to cause relaxation of cardiac stomach preparations from the starfish *A. rubens* (Cai et al., 2018). Therefore, from a functional perspective ArCT is more like AjCT1 than AjCT2 and we speculate that this may reflect loss of an exon encoding a AjCT2-like peptide in the Asteroidea lineage. Further insights into this issue and the evolution of CT-type neuropeptide function echinoderms could be obtained by investigation of the actions of CT-type peptides in brittle stars and sea urchins. Differences in the effects of AjCT1 and ArCT2 on the intestine may be explained by differences in their signal transduction mechanisms. Thus, only AjCT2, not AjCT1, triggered AjPDFR2-mediated activation of Ca²⁺ mobilization and ERK signaling and therefore we speculate that the relaxing effect of AjCT2 on the intestine may be mediated by activation of AjPDFR2.

3.4 AjCT2 stimulates feeding and growth in *A. japonicus*

In vivo pharmacological experiments revealed that treatment with AjCT2 at a high concentration (5×10^{-1} mg/ml) caused a significant increase in body mass, and this was accompanied by a decrease in residual bait and an increase in fecal mass. However, these effects were not observed with AjCT2 at a low concentration (5×10^{-3} mg/ml) or with AjCT1 at low or high concentrations. It is noteworthy that AjCT2, but not AjCT1, has a relaxing effect on the intestine in *A. japonicus* and therefore the effects of AjCT2 on feeding, defecation and growth may, at least in part, be mediated via binding of this peptide to receptors in the intestine. Accordingly, upregulation of the expression of the AjCT1/AjCT2 receptors AjCTR, AjPDFR1 and AjPDFR2 was observed in the intestine of animals injected with AjCT2 at a high concentration. Furthermore, changes in intestinal expression of genes encoding growth-related factors were also observed in animals injected with AjCT2 at a high concentration, with up-regulation of *AjIgf* and *AjMegf6* expression and down-regulation of *AjGDF-8* (Appendix 1—figure 1). Consistent with these changes in gene expression and the growth-promoting effect of AjCT2, it has been reported that *GDF-8* inhibits growth in *A. japonicus* (Li et al., 2016), whilst *Megf6* and *Igf* promote growth in vertebrates (Párrizas and LeRoith, 1997; Deribe et al., 2009).

Collectively, these findings indicate that AjCT2 has a physiological role in stimulating feeding and growth in *A. japonicus*. To the best of our knowledge, this is the first study to report orexigenic and growth-promoting effects of a CT-type neuropeptide in bilaterian animals. Interestingly, previous studies have reported the expression of CT-type neuropeptides in the digestive system of invertebrate chordates (Sekiguchi et al., 2009; Sekiguchi et al., 2016). However, in contrast to the stimulatory effect of AjCT2 on feeding/growth in *A. japonicus*, CT-type neuropeptides have been found to inhibit feeding and growth in vertebrates (Del et al., 2002; Hay et al., 2018; Nakamura et al., 2018). Thus, it appears that opposing roles of CT-type neuropeptide signaling systems as regulators for feeding/growth have evolved in deuterostomes, with a stimulatory role reported here in an echinoderm and an inhibitory role reported previously in vertebrates. Further research is now needed to investigate if the stimulatory effect of AjCT2 on feeding/growth observed in here in *A. japonicus* also applies to other sea cucumber species. Moreover, our discovery of this effect of AjCT2 on *A. japonicus* may provide a basis for development of novel approaches to improve aquaculture of this economically important species.

4. Materials and methods

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Cell line (Homo sapiens)	HEK293 cell line	Procell	CL-0005	
Antibody	Phospho-p44/42 MAPK ERK1/2(Thr202/Tyr204) (goat)	Cell Signaling Technology	CAT#4930	IF (1:2000)
Antibody	p44/42 MAPK ERK1/2 (137F5) (goat)	Cell Signaling Technology	CAT#4695T	IF (1:1000)
Antibody	HRP-conjugated goat anti-rabbit IgG (goat)	Absin	Abs20040ss	IF (1:2000)
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction--AjCTR	This paper	PCR PRIMER	FORWARD CGCGGATCCAAAATGCAGGAGA ACGATTC BamH I
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction--AjCTR	This paper	PCR PRIMER	REVERSE CCGGAATTCTTAAACTACCGTT GTTTTAT EcoR I
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction—AjPDFR1	This paper	PCR PRIMER	FORWARD CGCGGATCCACAATGAGTGTCTG GGAACGT BamH I
sequence-based reagent	Primers for pcDNA3.1(+)	This paper	PCR PRIMER	REVERSE CCGGAATTCTCATTTCCCACCCC

	plasmid construction— AjPDFR1			ACTCAC EcoR I
sequence- based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR2	This paper	PCR PRIMER	FORWARD CGCGGATCCAAGATGGCGAACA ATGAAAC BamH I
sequence- based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR2	This paper	PCR PRIMER	REVERSE CCGGAATTCTCATTCATTAAATT TGCCGT EcoR I
sequence- based reagent	Plasmid construction AjCTR–EGFP	This paper	PCR PRIMER	FORWARD CCGGAATTCTGAAAATGCAGGA GAACGATTC EcoR I
sequence- based reagent	Plasmid construction AjCTR–EGFP	This paper	PCR PRIMER	REVERSE CGCGGATCCCGAACTACCGTTG TTTTATCTG BamH I
sequence- based reagent	Plasmid construction AjPDFR1–EGFP			CCGGAATTCTGACAATGAGTGT CGGGAACGT EcoR I
sequence- based reagent	Plasmid construction AjPDFR1–EGFP			CGCGGATCCCGTTTCCCACCCC ACTCACTGT BamH I
sequence- based reagent	Plasmid construction AjPDFR2–EGFP			CCGGAATTCTGAAGATGGCGAA CAATGAAAC EcoR I
sequence- based reagent	Plasmid construction AjPDFR2–EGFP			CGCGGATCCCGTTCATTAAATTT GCCGTTCA BamH I
sequence- based reagent	Primers for qPCR assays β -Actin	This paper	PCR PRIMER	FORWARD AAGGTTATGCTCTTCCTCACGCT
sequence- based reagent	Primers for qPCR assays	This paper	PCR PRIMER	REVERSE GATGTCACGGACGATTTCACG

	β -Actin			
sequence-based reagent	Primers for qPCR assays β -Tubulin	This paper	PCR PRIMER	FORWARD GAAAGCCTTACGACGGAACA
sequence-based reagent	Primers for qPCR assays β -Tubulin	This paper	PCR PRIMER	REVERSE CACCACGTGGACTCAAAATG
sequence-based reagent	Primers for qPCR assays <i>AjCTP1/2</i>	This paper	PCR PRIMER	FORWARD GTTTGGCGGTACCCATTAC
sequence-based reagent	Primers for qPCR assays <i>AjCTP1/2</i>	This paper	PCR PRIMER	REVERSE GGCGTTCGGTCTGTTAATAG
sequence-based reagent	Primers for qPCR assays <i>AjMegf6</i>	This paper	PCR PRIMER	FORWARD CGGTCCTTACTGTGCGGAGA
sequence-based reagent	Primers for qPCR assays <i>AjMegf6</i>	This paper	PCR PRIMER	REVERSE CGTCCTCCCAGCTGGACATC
sequence-based reagent	Primers for qPCR assays <i>AjIgf</i>	This paper	PCR PRIMER	FORWARD TGGAAGGCAGTAAGGCTA
sequence-based reagent	Primers for qPCR assays <i>AjGDF-8</i>	This paper	PCR PRIMER	REVERSE TAGGCTGATGTGGTTGGA
sequence-based reagent	Primers for qPCR assays <i>AjGDF-8</i>	This paper	PCR PRIMER	FORWARD ATACGCCCATACTCACATCC
sequence-based reagent	Primers for qPCR assays <i>AjIgf</i>	This paper	PCR PRIMER	REVERSE TGGCAGGAGTCAACCTTCAT
sequence-based reagent	Primers for qPCR assays <i>AjCTR</i>	This paper	PCR PRIMER	FORWARD GACGAATGGAGGAGAAGG
sequence-based reagent	Primers for qPCR assays	This paper	PCR PRIMER	REVERSE CGCCGCAATTAGAGTAGT

	<i>AjCTR</i>			
sequence-based reagent	Primers for qPCR assays <i>AjPDFR1</i>	This paper	PCR PRIMER	FORWARD TTATCAACCGCCGTCTTC
sequence-based reagent	Primers for qPCR assays <i>AjPDFR1</i>	This paper	PCR PRIMER	REVERSE CAGGAGGAGGAACACATTC
sequence-based reagent	Primers for qPCR assays <i>AjPDFR2</i>	This paper	PCR PRIMER	FORWARD TGTCTCTCTGCCTCTTCT
sequence-based reagent	Primers for qPCR assays <i>AjPDFR2</i>	This paper	PCR PRIMER	REVERSE GTCTTCTCATCGGTGTG
sequence-based reagent	Primers for 3'RACE <i>AjCTP1</i> <i>AjCTP2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTCGAGCTG TGTCGATAACATTACGCATGG
sequence-based reagent	Primers for 3'RACE <i>AjPDFR1</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGAGGGAA AGCCAAACCAGCGAGACG
sequence-based reagent	Primers for 3'RACE <i>AjPDFR2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGTGCAAC CGGACGAAGATCCACGTG
sequence-based reagent	Primers for 5'RACE <i>AjPDFR1</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTCGTCTCG CTGGTTTGGCTTCCCTC
sequence-based reagent	Primers for 5'RACE <i>AjPDFR2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGCCTTCC ACAAACATCCATGCGAACGC
sequence-based reagent	Primers for 5'RACE <i>AjCTR</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTACGCCGG GCTCTCAAGAAGTCATTC
sequence-based reagent	Primers for in situ hybridization <i>AjCTP1/2</i>	This paper	PCR PRIMER	FORWARD CGAGCTGTGTCGATAACATTAC GCATGG

sequence-based reagent	Primers for in situ hybridization AjHOP	This paper	PCR PRIMER	REVERSE AATGGGTACCGCCAAACTGGCC TTC
sequence	RNAi	This paper	si <i>AjCTP1</i> -1	GGAAUUAUCAAGACAGUAU (dT)(dT) AUACUGUCUUGAUAAUCC (dT)(dT)
sequence	RNAi	This paper	si <i>AjCTP1</i> -2	GAAAGUCGCUAAUGCAGUA (dT)(dT) UACUGCAUUAGCGACUUUC (dT)(dT)
sequence	RNAi	This paper	siNC	UUCUCCGAACGUGUCACGU (dT)(dT) ACGUGACACGUUCGGAGAA (dT)(dT)
commercial assay or kit	SMARTer RACE 5'/3' Kit	Takara	Cat No. 634858	
commercial assay or kit	FastPure Gel DNA Extraction Mini Kit	Vazyme	DC301-01	
commercial assay or kit	Hifair® III 1st Strand cDNA Synthesis SuperMix	YEASEN	Cat No. 11141ES60	
commercial assay or kit	Hieff UNICON Universal Blue qPCR SYBR Green Master Mix	YEASEN	Cat No. 11184ES08	
commercial assay or kit	DIG RNA Labeling Kit (SP6/T7)	Roche	Cat No.1117502 5910	
commercial assay or kit	DIG Nucleic Acid Detection Kit	Roche	Cat No.1117504 1910	
commercial assay or kit	Firefly luciferase reporter gene assay kit	MK	MF4001	
commercial assay or kit	cAMP Assay kit	R&D systems	Cat No. KGE002B	

4.1 Animals

Adult sea cucumbers were collected from an aquaculture breeding farm in Weihai, Shandong, China. All the animals were acclimated in breeding aquaria (18 °C, 32 ppt) for at least seven days and fed daily with a commercially formulated diet. Tissues used for cDNA cloning and qRT-PCR were obtained from five adult *A. japonicus* (~120 g) and kept at -80 °C until use. Circumoral nervous tissues from three adult sea cucumbers (~120 g) were fixed with 4% paraformaldehyde (PFA, G-CLONE, Beijing, China) and used for *in situ* hybridization analysis. *In vitro* functional analysis and *in vivo* experiments were carried out using twelve (~120 g body weight) and 168 (~20 g) sea cucumbers, respectively.

Ethical approval

All methods were carried out in accordance with the ARRIVE guidelines (<https://arriveguidelines.org>) for animal experiments. All animal care and use procedures were approved by the Institutional Animal Care and Use Committee of Ocean University of China (Permit Number: 20141201). They were performed according to the Chinese Guidelines for the Care and Use of Laboratory Animals (GB/T 35892-2018).

4.2 Molecular structure and phylogenetic analysis of CT-type neuropeptides and precursors

A. japonicus CT-type neuropeptide precursor sequences (*AjCTP1*, GenBank accession number: MF401985; *AjCTP2*, GenBank accession number: MF401986) were identified in our previous study (Chen et al., 2019). Based on the published sequences, transcript-specific primers for *AjCTP1* and *AjCTP2* (Supplementary Table 1) were designed using Primer3Plus (<https://www.primer3plus.com/>) and sequence accuracy was confirmed by SMARTer RACE 5'/3' Kit (TaKaRa, Kusatsu, Japan) according to the manufacturer's instruction. A sequence translation tool (<http://www.bio-soft.net/sms/index.html>) was used to obtain the corresponding amino acid sequences encoded by the cloned cDNAs. To identify the conserved features of CT-type neuropeptides in Bilateria, sequences were aligned using Clustal-X with default settings and then predicted using the online tool Multiple Sequence Comparison Display Tool (<http://www.bio-soft.net/sms/index.html>). Tertiary structures were investigated using SWISS-MODEL (<https://swissmodel.expasy.org/interactive>). A neighbor joining tree was constructed to investigate relationships of *AjCTP1*, *AjCTP2* and CT-type precursors from other Bilateria. Amino acid sequences were aligned using MEGA7 (v.7.170509) with default settings and then modified by iTOL (<https://itol.embl.de/login.cgi>). The accession numbers or citations for all sequences used for analysis are listed in Supplementary Table 2.

4.3 Cloning, sequence analysis and phylogenetic tree construction for candidate AjCT1/2 receptors in *A. japonicus*

Five potential receptors for AjCT1 and AjCT2 in *A. japonicus* (GenBank accession number: PIK49998.1, PIK49999.1, PIK49604.1, PIK34007.1 and PIK62458.1) were identified based on genome-wide analysis (Huang et al., 2021). Sequence comparison revealed that both PIK49998.1 and PIK62458.1 were partial sequences of PIK34007.1. Thus, three candidate receptors named as *AjCTR* (Calcitonin receptor, GenBank accession number: PIK49604.1), *AjPDFR1* (Pigment-dispersing factor receptor 1, GenBank accession number: PIK34007.1), and *AjPDFR2* (Pigment-dispersing factor receptor 2, GenBank accession number: PIK49999.1) were predicted and then analyzed. To confirm the accuracy of these sequences, cDNAs were cloned using the SMARTer RACE 5'/3' Kit (Takara, Cat No. 634858) and then sequenced (Beijing Genomics Institution, China). Receptor sequence analysis and phylogenetic analysis were performed as described in 4.2. In addition, the online tool NetPhos-3.1 (<https://services.healthtech.dtu.dk/service.php?NetPhos-3.1>) was used to predict potential phosphorylation sites. The conserved domains of protein sequences were predicted by the Batch SMART plugin of TBtools (Toolbox for biologists, v.1.098652) software. For

analysis of gene synteny, neighboring genes of *AjCTR*, *AjPDR1* and *AjPDR2* were found based on an *A. japonicus* chromosome-level genome (Wang et al., 2022) and the corresponding genes in the starfish *Asterias rubens* and the sea urchin *Lytechinus variegatus* were also identified by performing BLASTp in the NCBI database. Specific primers for cloning of cDNAs encoding the candidate receptors are shown in Supplementary Table 1 and the details of the sequences used for constructing the phylogenetic tree are listed in Supplementary Table 3.

4.4 Testing *AjCT1* and *AjCT2* as ligands for candidate receptors and exploration of downstream signaling pathways

AjCT1 and *AjCT2* were custom synthesized by GL Biochem Ltd. (Shanghai, China). Subsequently, CRE (cAMP response element) -Luc (Luciferase) detection, cAMP accumulation, SRE (serum response element) -Luc detection, receptor translocation assays, and Western blotting were carried out.

AjCTR, *AjPDR1* and *AjPDR2* incorporating enhanced green fluorescent protein (EGFP) were constructed respectively according to the method described by Li et al. (2022). Briefly, the complete ORFs of candidate receptors were cloned and sequenced. Then, the PCR products were digested with EcoRI and BamHI (TaKaRa, Kusatsu, Japan) and then incorporated into the pEGFP-N1 vector (YouBio, Changsha, China) that was subject to the same double-digestion. Finally, the constructed plasmid with correct sequences was transfected into HEK293T cells seeded on glass coverslips in 12-well plates. After incubation with *AjCT1* (10^{-6} M), *AjCT2* (10^{-6} M) or serum-free DMEM (Biosharp, Hefei, China) for 15 min, cells were washed with PBS (Biosharp, Hefei, China) and fixed with 4% PFA for 10 min. Finally, cell nuclei were stained with DAPI (G-CLONE, Beijing, China) for 5 min and an Olympus bright field light microscope was used for image capture. The pcDNA 3.1(+) vectors (YouBio, Changsha, China) comprising the complete ORF of the three candidate receptors were also constructed as described by Li et al. (2022). Then, they were co-transfected with the cAMP response element, pCRE-Luc (YouBio, Changsha, China) or the pSRE-Luc plasmid including PKC reporter gene (YouBio, Changsha, China) into HEK293T cells. Incubation was carried out with *AjCT1* (10^{-6} M), *AjCT2* (10^{-6} M) or serum-free DMEM (negative control) for 4 h at 37 °C. Finally, luciferase activity was tested using the firefly luciferase assay kit (MK, Shanghai, China). When required, transfected cells described above were co-incubated with H89 (PKA inhibitor) (Absin, Shanghai, China), FR900359 (Gaq protein inhibitor) (Cayman Chemical, Ann Arbor, Michigan) or Gö 6983 (PKC inhibitor) (Cayman Chemical, Ann Arbor, Michigan) before the experiment to explore the potential signaling pathways. To confirm the concentration of intracellular cAMP after stimulation, HEK293T cells expressing pcDNA 3.1(+)/*AjCTR*, pcDNA 3.1(+)/*AjPDR2* or pcDNA 3.1(+)/*AjPDR1* were treated with *AjCT1* or *AjCT2* (10^{-9} – 10^{-5} M) respectively for 15 min after 2 h starvation in serum-free medium. Subsequently, cAMP was measured using the ELISA-based cAMP assay kit (R&D systems, USA&Canada) following the manufacturer's instruction. HEK293T cells transfected with empty pcDNA3.1(+) were used as a negative control. The specific primers used for constructing plasmids are listed in Supplemental Table 1.

Western blotting was performed as described previously (Li et al., 2022). Briefly, HEK293T cells transfected with the constructed pcDNA 3.1(+) plasmid were starved in serum-free medium for 2 h, and then incubated with *AjCT1* or *AjCT2* (10^{-6} M). PVDF membranes containing proteins were blocked with 5% BSA (G-CLONE, Beijing, China) and incubated overnight with phosphorylated ERK1/2 kinases (CST, Boston, China). After that, goat anti-rabbit horseradish peroxidase-conjugated secondary antibody (Absin, Shanghai, China) was applied and incubated, followed by the addition of Omni-ECL™ Femto Light Chemiluminescence reagents (EpiZyme, Shanghai, China). Finally, RVL-100-G (ECHO, America) was applied for visualization of the PVDF membrane. When required, transfected cells were co-incubated with PKA or PKC inhibitor before the experiment to identify signaling pathways that activate downstream ERK1/2 cascade.

4.5 Total RNA extraction and Quantitative

Real-Time PCR (qRT-PCR) detection

qRT-PCR was employed to detect *AjCTP1* and *AjCTP2* transcript expression in five tissues from *A. japonicus*, including the intestine, longitudinal muscle, respiratory tree, gonad and circumoral nervous system. Because it was difficult to design specific primers for *AjCTP1* and *AjCTP2* (Zheng et al., 2022 [\[4\]](#)), total transcript expression levels for *AjCTP1/2* (*AjCTP1* + *AjCTP2*) were investigated. Total RNA was extracted from the collected tissues using Trizol RNA isolation reagent (Vazyme, Nanjing, China), cDNA was synthesized using Hifair® III 1st Strand cDNA Synthesis SuperMix (YEASEN, Shanghai, China) according to manufacturer's instructions. For qRT-PCR, amplification was performed in 20 µL volume reactions based on Hieff UNICON Universal Blue qPCR SYBR Green Master Mix (YEASEN, Shanghai, China) and expression levels were assessed by a Corbett Rotor-Gene Q (Qiagen, Germany). β -actin (GenBank accession number: PIK61412.1) and β -tubulin (GenBank accession number: PIK51093) were used as house-keeping genes for normalization (Zhao et al., 2014 [\[4\]](#)). The $2^{-\Delta\Delta CT}$ method was applied to analyze the relative expression levels and comparison between tissues was carried out using one-way analysis of variance (ANOVA) followed by a Tukey post hoc test (IBM SPSS 25.0 software, Chicago, IL, USA). A p value < 0.05 was considered as a statistically significant difference. Specific sequences of primers used for qRT-PCR are listed in Supplemental Table 1.

4.6 Localization of *AjCTP1/2* expression in

A. japonicus using in situ hybridization

Digoxigenin-labeled RNA antisense probes and corresponding sense probes were synthesized using DIG RNA Labelling Kit (Roche, Basel, Switzerland) and the specific primers are listed in Supplementary Table 1. Then, localization of the total *AjCTP1/2* transcripts was detected in the circumoral nervous system. The detailed method used was described by Li et al. (2022) [\[4\]](#). Briefly, the tissue was prehybridized after rehydration and proteinase K treatment, following the addition of antisense RNA probes that bind specifically to mRNA (sense RNA probes were used for negative control). Blocking buffer was used for blocking non-specific binding sites and then alkaline phosphatase-conjugated anti-digoxigenin antibodies were added according to the manufacturer's instruction, which could specifically bind to the hybridized nucleic acid probe. Finally, the chromogenic substrate NBT/BCIP was added for visualization of *AjCTP1/2* expression. Images were taken using an Olympus bright field light microscope (Olympus, Tokyo, Japan).

4.7 Analysis of the in vitro activity of *AjCT1*

and *AjCT2* on preparations from *A. japonicus*

Longitudinal muscles and intestines (20 mm in length) from *A. japonicus* were dissected and then suspended in a 50 mL glass organ bath containing 20 mL sterilized seawater. Both ends of the preparations were tied with cotton ligatures and then one end was tied to a metal hook and the other end was connected with a High Grade Isotonic Transducer (ADInstruments MLT0015, Oxford, UK). Before testing, the preparation was stabilized for about 20 min, and the resting tension was adjusted to 1.0 g. Informed by previous findings from Cai et al. (2018) [\[4\]](#), we investigated if *AjCT1* and *AjCT2* caused relaxation of preparations. First, acetylcholine (ACh) was applied to induce the contraction of preparations and then when a stable state of contraction was achieved, 20 µL *AjCT1* or *AjCT2* at a range of concentrations was added and relaxation responses were recorded. Output signal (cm) from PowerLab data acquisition hardware (ADInstruments PowerLab 4/26, Oxford, UK) was calculated using LabChart (v8.0.7) software.

4.8 Investigation of the in vivo pharmacological effects of AjCT1 and AjCT2 in *A. japonicus*

A total of 150 adult sea cucumbers were randomly assigned to five groups (30 individuals per group), including two high concentration groups (CT1H and CT2H, 5×10^{-1} mg/ml), two low concentration groups (CT1L and CT2L, 5×10^{-3} mg/ml) and the control group (CO, sterilized seawater), with each group consisting of three parallel experiments comprising ten animals. Before the experiments, sea cucumbers were starved for a day to normalize their physiological status and their wet weights were measured. AjCT1 and AjCT2 were dissolved in sterilized seawater and then injected through the peristome at noon every other day (1 μ l/g wet weight) and the CO group were injected in the same way but with sterilized seawater (1 μ l/g wet weight). Faeces and residual bait were collected and weighed after drying every day. After 24 days, wet weights of sea cucumbers were measured again. Subsequently, animals were dissected and intestines were collected for detecting the expression levels of the three receptor genes (*AjCTR*, *AjPDR1*, *AjPDR2*) and several growth factors, including multiple epidermal growth factor 6 (*AjMegf6*, GenBank accession number: MG018199.1), insulin-like growth factor (*AjIgf*, GenBank accession number: PIK50518.1) and growth and differentiation factor-8 (*AjGDF-8*, GenBank accession number: KP100064.1) (Li et al., 2016; Wang et al., 2018). IBM SPSS Statistics 25.0 was applied for statistical analysis and a *p*-value < 0.05 was considered as significant difference. The specific primers for qRT-PCR analysis are listed in Supplementary Table 1.

4.9 Loss-of-function experiments by RNA interference (RNAi)

To investigate how receptor expression is affected by knock-down of ArCTP expression, an RNAi experiment was carried out. It was difficult to design *AjCTP1*- and *AjCTP2*- specific interference oligonucleotides and therefore, we designed and customized generic interference oligonucleotides targeting both *AjCTP1* and *AjCTP2* (si*AjCTP1/2-1* and si*AjCTP1/2-2*, containing the extra 2' ome modification) (Beijing Tsingke Biotech Co., Ltd. China). Meanwhile, a negative control siRNA (siNC) was also customized for the experiment (Supplementary Table 1). The si*AjCTP1/2-1*, si*AjCTP1/2-2* or siNC dry powder was diluted to 20 μ M with DEPC water following the manufacturer's instruction, then 60 μ l sterilized seawater was mixed with 20 μ l si*AjCTP1/2-1*, si*AjCTP1/2-2* or siNC, and 20 μ l Lipo6000 transfection reagent (Beyotime, Shanghai, China) as the transfection solution. Eighteen sea cucumbers were randomly divided into three groups and injected with three different transfection solutions respectively. The mixed solutions were injected through the peristome once a day (1 μ l/g wet weight) and intestines were dissected and frozen in liquid nitrogen 3 days after injection. The expression level of *AjCTP1/2* was investigated by qRT-PCR and the interference rate was calculated to determine the inhibition efficiency of siRNAs. Finally, the relative expression levels of *AjCTR*, *AjPDR1* and *AjPDR2* were investigated using the selected *A. japonicus* intestines.

Competing Interests

The authors declare that no competing interests exist.

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Author Contributions

Xiao Cong, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing—original draft preparation; Huachen Liu, Methodology, Validation; Lihua Liu, Methodology, Validation; Maurice R. Elphick, Formal analysis, Validation, Writing—review & editing; Muyan Chen, Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing—review & editing

Supplementary figures

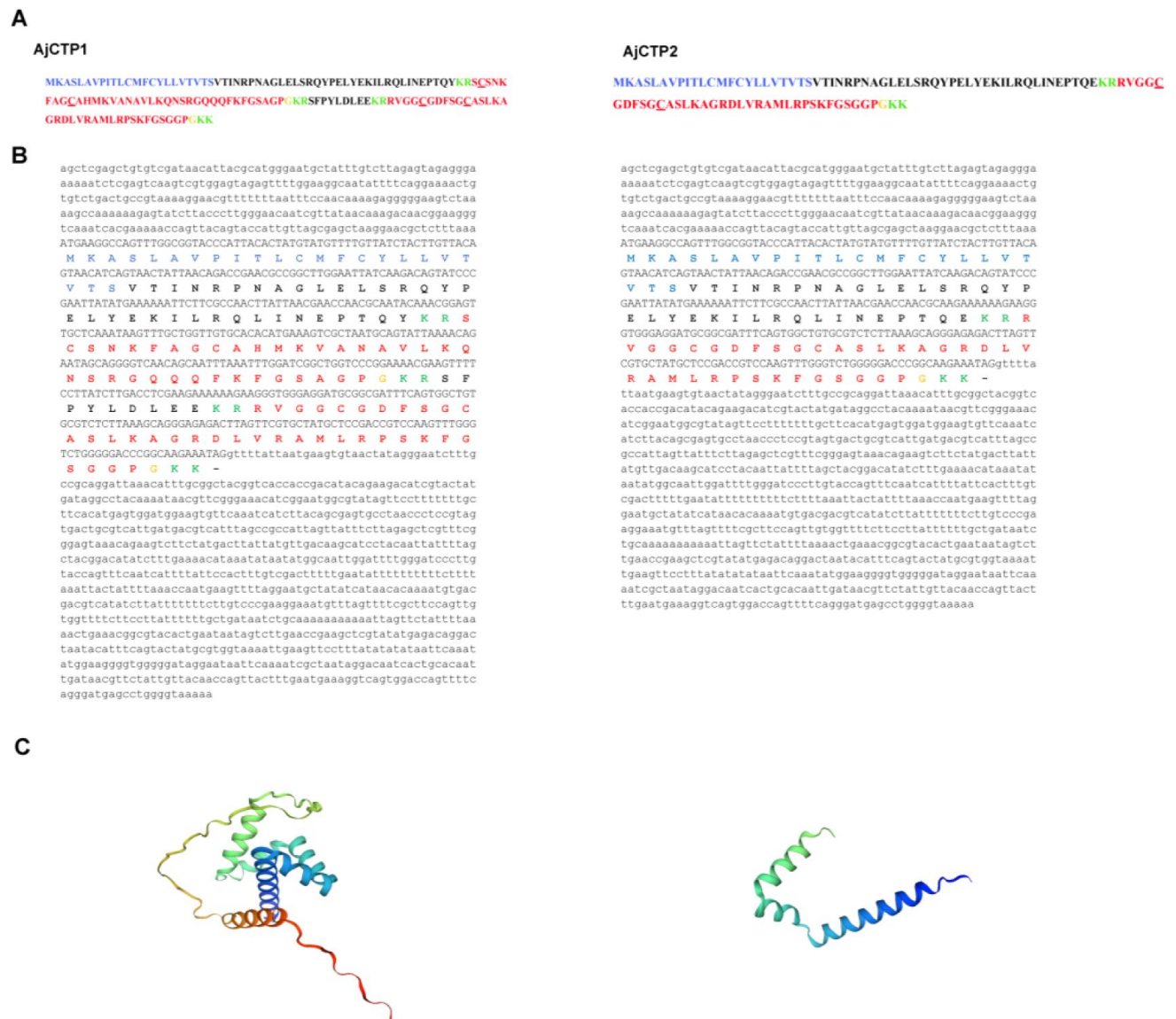


Figure 1-figure supplement 1.

Calcitonin-type (CT-type) neuropeptide precursors in *A. japonicus*. (A) *AjCTP1* comprises two putative CT-type neuropeptides, AjCT1 and AjCT2, whereas *AjCTP1* generated by alternative splicing comprises only AjCT2. The predicted signal peptide is shown in blue, the predicted cleavage sites

(KR or KK) are shown in green, CT-type neuropeptides are shown in red and C-terminal glycines that are potential substrates for amidation are shown in orange. Predicted disulfide bridges in AjCT1 and AjCT2 are formed between the two underlined cysteines. (B) Nucleotide sequences encoding CT-type neuropeptide precursors and the deduced amino acid sequences in *A. japonicus*. 3'UTR and 5'UTR were shown in lowercase and ORF was shown in uppercase. Blue uppercase labeled the predicted signal peptide, the green uppercase showed the predicted cleavage site, AjCT1 and AjCT2 were shown in red and the putative substrate for amidation was shown in orange. (C) The tertiary structures of AjCTP1 and AjCTP2 as predicted by SWISS-MODEL

A_japCT1	----	SCSNKFA	G	-	CAHMKVA	NAV	L	K	Q	N	S	R	G	Q	Q	F	K	F	-	-	GSAG	-	Pa										
A_japCT2	--RVGG	CGD	-	FSG	-	CASLKA	G	R	D	L	V	R	A	M	L	R	-	-	PSKF	-	-	GSGG	-	Pa									
H_scaCT1	----	SCSDRF	S	G	-	CAHLKVA	K	A	L	L	D	Q	-	A	R	R	E	N	S	R	F	-	Pa										
H_scaCT2	--	MGG	C	G	D	-	FSG	-	CASLKA	G	R	D	L	V	R	A	M	L	R	Q	-	-	Pa										
H_glaCT1	----	SCSDRF	A	G	-	CTHLKLA	K	V	L	I	D	Q	-	S	R	R	D	G	N	I	R	F	-	Pa									
H_glaCT2	--	MGG	C	G	D	K	F	S	G	-	CASLKA	G	R	D	L	V	R	A	M	L	R	Q	-	Pa									
S_purCT	--	SKG	C	G	S	-	FSG	-	CMQMEVA	K	N	R	V	A	A	L	L	R	N	S	N	A	H	L	F	-	Pa						
A_rubCT	NGESRG	CSG	-	FGG	-	CGVLT	I	G	H	N	A	A	M	R	M	L	A	E	S	-	N	S	P	F	-	Pa							
A_plaCT	---	SSSGCAEY	F	G	G	-	CAQLKL	G	Q	D	A	L	S	R	M	L	A	D	S	-	N	S	R	F	-	Pa							
P_pecCT	---	SGTGCTQ	-	FSG	-	CAQLKV	G	Q	D	A	L	S	R	V	L	A	D	S	-	N	S	R	F	-	Pa								
O_vicCT1	SGNGGC	A	G	-	FTG	-	CAQLAAG	Q	N	A	L	R	N	F	M	H	S	N	R	A	S	L	F	T	-	Pa							
O_vicCT2	NGNGGC	A	G	-	FTG	-	CAQLAAG	Q	S	A	L	Q	A	M	I	H	S	G	R	A	S	L	F	-	Pa								
O_araCT1	GTEKGC	S	G	-	FSG	-	CAQLAAG	Q	S	A	L	Q	A	M	I	H	G	N	R	A	S	L	F	-	Pa								
O_araCT2	GGGGCKS	-	FSG	-	CAQLVI	G	Q	N	A	V	R	N	M	M	H	S	N	R	A	S	I	F	S	-	Pa								
A_filCT1	GTEKGC	A	G	-	FSG	-	CAQLAAG	Q	S	A	L	Q	A	M	I	H	G	N	R	A	S	L	F	-	Pa								
A_filCT2	GGGCRG	-	FSA	-	CAQLAI	G	Q	D	A	F	R	N	M	M	H	N	N	R	A	S	I	F	T	-	Pa								
A_medCT	---	G	-	C	-	DVFGG	-	CAQLKV	G	R	E	L	A	I	D	T	L	Q	K	G	S	G	G	M	F	-	Pa						
An_japCT	---	GDC	-	DSFGG	-	CAQLTAS	Q	Q	L	A	Y	R	T	R	H	R	D	L	S	N	L	F	-	Pa									
S_kowCT1	---	GEGCKG	-	FAA	-	CGNLDA	G	R	K	H	W	K	A	N	M	G	G	G	R	T	A	M	E	S	S	A	G	T	-	Pa			
S_kowCT2	GRGTSAC	G	G	-	FAT	-	CKQLEY	G	R	K	Y	A	T	S	-	-	K	A	D	L	S	H	F	-	G	A	T	S	-	Pa			
B_floCT1	--GK	I	A	C	K	T	-	AW	-	CMNNRL	S	H	N	L	S	S	L	D	N	P	T	D	T	G	V	G	-	-	-	-	Pa		
B_floCT2	----	DCST	-	L	T	-	CFNQKL	A	H	E	L	A	M	D	N	Q	R	T	D	T	A	N	P	Y	-	-	-	-	-	-	Pa		
B_floCT3	----	KCES	-	G	T	-	CVQMHL	A	D	R	L	R	L	G	L	G	H	N	M	F	T	N	T	-	-	G	P	E	S	-	Pa		
C_intCT	----	CDG	-	V	S	T	-	CWLHEL	G	N	S	V	H	A	T	A	G	G	K	-	N	V	G	F	-	-	-	-	-	-	Pa		
H_sapCGRP	----	ACDT	-	A	T	-	CVTHRL	A	G	L	L	S	R	S	G	G	V	V	K	N	N	F	V	P	T	-	N	V	G	S	K	A	Fa
H_sapCT	----	CGN	-	L	S	T	-	CMLGTY	T	Q	D	F	N	K	F	H	T	F	P	-	Q	T	A	I	-	-	G	V	G	A	-	Pa	
L_gigCT	----	SCSLRL	G	M	C	L	T	E	N	L	N	A	A	N	Q	Y	-	-	-	-	-	E	Y	L	S	-	S	G	L	S	-	Pa	
D_melDH31	TVDFGL	A	R	G	Y	S	G	-	-	TQEAKHR	M	G	L	A	-	-	-	-	-	-	A	A	N	F	-	-	-	-	-	-	Pa		
C_telDH31	----	RFDAGY	G	S	-	-	RYGVA	Q	S	V	G	S	K	L	M	A	L	K	Q	A	A	D	W	-	-	-	-	-	-	-	Pa		
C_telCT	----	TCQFNL	G	G	H	C	A	T	E	S	A	A	S	V	A	D	H	W	-	-	-	-	-	-	-	-	-	-	-	-	Pa		

Figure 1-figure supplement 2.

Comparison of CT-type neuropeptides from Bilateria: deuterostomes-ambulacraria (green), chordata (purple) and protostomes (pink). The conserved residues are shown with white lettering highlighted in gray and black. The accession numbers or citations are shown in the Supplementary Table 2.

A

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1  cagaatcctttcacccgacctctgagctgaatcaaatcgaattcaggagtgtggtgat
1  M Q E N D
61  gtcaaaacggacactggcagtcacgaagatcgaaaATCGAGGAGAACGATCATATT

9  F S F G L N D P C E E F E D F D
121 ATTCAGCTTTGGTTCCTTTACAGACCCGAGTTGCTCTGAAGAATTGAAGACTTTG

29  P L A A D Q I D E W R R R R D C C E
181 ACCCGCTTGCTACAGCTGACCAATAGACGAATGGAGGAGAAGCGAGATTCTTGTGCG

49  R I E N E P P D E P H C L V W B D T W
241 AAAGAATTGAAACGAGAGCCCTCTGACGAGCCGATTGACGCTTGTATGGGATACGT

69  D C W N N W T P G D L V Q P C P W W V
301 GGGATTGTTGGAATGGACTAGACCCGAGATACCTTGTTCAACCATGCCATGGTGGG

89  P Q V N P N K D A M K C M P D G W F
361 TTCCACAAGTTAATCCGAACAAAGATGCCATGAAACATGCATGCTGATGGGACGTGGT

109 K H P D T N T W N N C G A Q S
421 TCAAACTCCCGCACTAATATACCTGGACCACTACTCTAATTGGGGCCATCAGAGA

129 I H V A I F Y T G Y V V I S L C L
481 GCACATATCATGTGGCCATATCTACACGGATACAGTGTTCAGTTATATCACTCTGCT

149 A L F I F Y F Q S L G C P R V I H K
541 TAGCCCTCTCATATTACATACCTTTAGAGCCTTGGATGCCGAGGGTTACCATCCACA

169 N L F I S F I L C I S N I T W H I T I
601 AAAATCTGTTTCATCTCTCTATCTTGAGCTGCATATCAAACTACGCTGGCATATCACA

189 I A S A E K Q E T A C R I H I I A Q
661 TCATTGCTATCAGCGGAGAGCAACAGGAGACGGCATGTGCAATAACTCATATCATTCGCG

209 F F V L C N Y F W M L E G L L H V
721 AGTTTTTTGCTTTGCAACTACTTTTGATGCTAAGTGAGGGATTATACCTACATACGG

229 I V V A V F S E N H N L L L Y Y I V G W
781 TGATAGTGGTTGCGAGCTTCTCGGAAATCATATCTCTCTCTATTATATCTGTCGGAT

249 V I F L V F A V F L S L L L T Q D G G
841 GGGTGATTCATTGGTCCGACGAGTGTTCCTCAGCTTGTATTAAACAGAGATGGTG

269 R C W N A N E I E W V V G G I I A V
901 GAAGGTGTTGGACGAACGCTAATGAGATCGAATGGGTCTGGGTGGAACTATTATTGCGAG

289 L L T N A A L L L N I V R V L V K L R
961 TATTATTGACAAACGCTGCTCTCTCTCTTAACATAGTACGCGTTTATGTCACGAAGCTTA

309 A T P S Q G A R T Y V R A V R A T I I L
1021 GGGCTACCAAGGTGCAAGAACATACGTAAGAGCTGTCGAGCTCATTATTTC

329 L P L M G L H I I V P V R P D N H V
1081 TGCTTCCTCTGATGGGCTCCATTATATCATCGTCCAGTGAGGCCAAGTGATAAACATG

349 A E V I D C I V A F L I S F Q G L F V
1141 TAGCCGAAGTTATTACGATTGTATCGTCGCTTTCTTAATTCATTTCAGGGCCTTTTCG

369 A C I F C F F N G E V K M Q I R R K W L
1201 TAGCCTGCATATTTGTTTCTTCAATGGCGAGTAAAGTGCAGATCAGGAGAAAATGGC

389 N W Q F R R G D F R S R H V
1261 TATCCAACCTGGCAGTTGAGAAGTGGTGACTTCGGTCACGACACCACTACTACTG

409 E A V T P P L V N L H E K D L
1321 TAACCGAAGCTGTATCCACTACACCACTCTGAGTGTCAATCTTCATTCGGAGAAAGATT

429 D N I N H P I K K N G I N G D N G K
1381 TGGTAACATACATCAATCATCTATCAAGACAAAGATGGGATTAACGGGGCAATGGGA

449 K K V G F T E A T I A N D M E R K
1441 AGAGTAAAAATCGGTGGATTTCAGAGGCGACGATTGCTAACGACATGGAAGTCGAA

469 F L M G K D A A E D P D K T V V *
1501 AACCATTTGATGACAGGGAAGATGCGGCCGAAGATCCAGATAAAACACGGTAGTTTAAa
1561 atgggaggaaggaaggaatgaatgaatgactctcttgagagccgcgcttcatctctggga
1621 tgtaaaaggggacggatggatccccatatactacattatgtgtactctacaactt
1681 gcttcggagcgacactcttgacatctagacatcggttactactattgcctcaaatgttg
1741 ccaaatgacgcaagaagaagaacacatcacagagcttttagaaaattgattttccattc
1801 tgcgaattttgacatacaagttacagtatctgtgttgctctgcactgtctatgtacca
1861 acccagtttggtaacagctctttttagctctgttttgcctgcaatgtgtgtgaaat
1921 agttatcacgctctcctcaacagctcttaacaatttaacgtttagaaaatgatgaacaggac
1981 ttacatttgacaatactataaaattactgtgtcgaatgtgaattctgttttccatttg
2041 gttttaagggttattctgtatactacggcagatacattccaacgtagtttgggaaga
2101 ctacgataacttgaggttagatgaagcggaattgatctagctcgttttgcgtggaactaga
2161 gatttgaagaaaaaaatcattatattatgtgtatcgtaaatgtgaagcaagctct
2221 cacttacattataactctcagatatgatccagctcatgcaagaacagaggttagatagaa
2281 agtttagagcatggaagtatttttagctgcataatttatagttcaagaataaagccaaaa
2341 ctgaagtaagtgatgtttccaaagtgatgagattttcgtgatattcagctagaaaaca
2401 tgcataaactctggaatgctcctttggggttaacttgatgatgaggtctgaacctcagc
2461 gtggcctaaatgtattgttaatacaatgttccaaatttatgaagtatacctatatat
2521 atatgtttttgtatatatatatacaoggtttatatatatatatata

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Figure 1-figure supplement 3.

Amino acid sequences and phosphorylation sites of AjCTR, AjPDFR1 and AjPDFR2 in *A. japonicus*. (A), (B) and (C) show the cDNA sequences encoding the receptors and their deduced amino acid sequences. ORFs are shown in uppercase; 3'UTR and 5'UTR are shown in lowercase. The stop codon is shown by the asterisk (*). Predicted potential phosphorylation sites are shown with squares.

B

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1 agagagagaggtggagagagagagagtattcttatcagccaatcgggtgtaatttcaat
61 cegttaaagtgtgcgcgtctctttccgattccctatagaagtcagttggttgatggga
121 acataccggtagctgggttcctatgagaacccattgcacaagatctgttaagcctacagtcg
181 gttgtctcgttagtaagtttacccgcatgatactgtcatattaaactcattagttgat
241 agctatggtagtgcattacaaagtaacttgaacaaatctgcctcagttggttgg
301 tatttttcattattctttcaggagcattcttttgaattgtaaaactaaacggtttggg
361 ttgtgctttggcagtgagctttaagttgtctgttactacgattgatcatatttccacc
421 gtaactcaacaatatataactgcgcaacaATGAGTGTGGGAACTGACCTGCCCTGACA

12 S R [ ] [ ] I I P Y D V T G M T L F C P E T
481 ATTCCAGGAGCAGTATTATCTCTATGACGTACAGGTATGACTTTGTCTGTCCCGAGA

32 [ ] D [ ] [ ] T L C W P D [ ] P P N T T V H I D C
541 CCTATGACTTACTCTCTCTGCGCAGATACACCACCGAACGACAGCATATAGATT

52 P I F P W N A G I D I D A Q A H R R C G
601 GCCCGATTTTCCATGGAACGCTGGAATTGACATTGATGCCAGGCACATGTCGGTGTG

72 A D G F W E Q H V [ ] E D C [ ] Q I G P P G
661 GGGCTGATGGATTCTGGGAACAGCATGTCTACGAAGATTGCTCGCAGATAGGACACCTG

92 T V P P C C G [ ] [ ] E D D L K F Y [ ] K V I N G
721 GAACCTTCTCTCTGGAATACAGAGGATGATCTTAAGTTTATAGCAAGTAATAATG

112 A K I L E V I G I T M [ ] L M A V L V A L
781 GTGCCAGATCCTGGAGGTGATTGGGATAACTATGCTCTTATGGCGGTGCTGGTAGCAT

132 Y I F H [ ] F R [ ] L R N H R [ ] R I H Q H L
841 TGTACATATCCATTCTTTCCAGGAGTTTGGCAATCATCTACTCGAATACATCAACATC

152 F L A F L I R L [ ] L D V I F M I N R F K
901 TGTTCCTAGCCTCTCTATCCGACTCACTCTCGATGTTATATTCATGATTACAGGTTCA

172 K K A [ ] D [ ] L E P V G I N K F F F L V K
961 AAAAGAAAGCCAGATTCTTTGGAACAGTTGGCATCAACAAATTTCCACATTGGTGA

192 V M E L C R E Y A R L C [ ] F T W M F V E
1021 AAGTAATGGAACTTTCCGGAATATGCCGATTGTGTACGTTTACGTGGATGTTGTGG

212 G I [ ] L N [ ] L L [ ] [ ] A V F R K P N F L W
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232 Y Y L I G W A S P I P F V L A F C I A M
1141 GGTACTATCTCATTTGGCTGGGCTCACCAGATACATTGTGCTAGCATTTTGCATTGCGA

252 Q K T S [ ] N G [ ] W H I [ ] T E S N Y [ ] V F
1201 TGCAAAAGACATCTTCAAATGGCTATTGGCATATATACATGAATCAAATTATTACGTAT
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C

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1 M A N
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4 N E T S V L C L E I C K D [ ] D Y D V V C
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24 W P E T A N G T I S N Q P C P D [ ] P F T
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44 D P [ ] Q L F E R L C Q P D G Q W E D V F
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1261 TTTTCATTGAGGGACCAAGAAACATTATCGTCGTGATGAATGTGTTCTCTCCTCTGAACA

292 I R V L I [ ] K L R E [ ] Q T [ ] E [ ] K Q V R
1321 TTATCAGAGTCTGTATACAAAACCTGAGGAAAGCCAAACCGCGAGACGAAACAGTCA

312 K A [ ] K A C I V L L P L L G T A N L V W
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332 I P K V P E P G K [ ] [ ] R F Y F A S Y Y Y
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352 V V L F L D A Y Q G F F L A L L Y C F L
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372 N I D V R V [ ] I K R K W Q G W M N A R N
1561 TAAACATTGATGCCAGTTACGATAAAAAAGAAATGGCAAGGTTGGATGAACGCCAGAA

392 P Y R A H G [ ] V V T [ ] [ ] T D V R N [ ] [ ] L
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412 D N N D [ ] E W G G K *
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Figure 1-figure supplement 3. (continued)

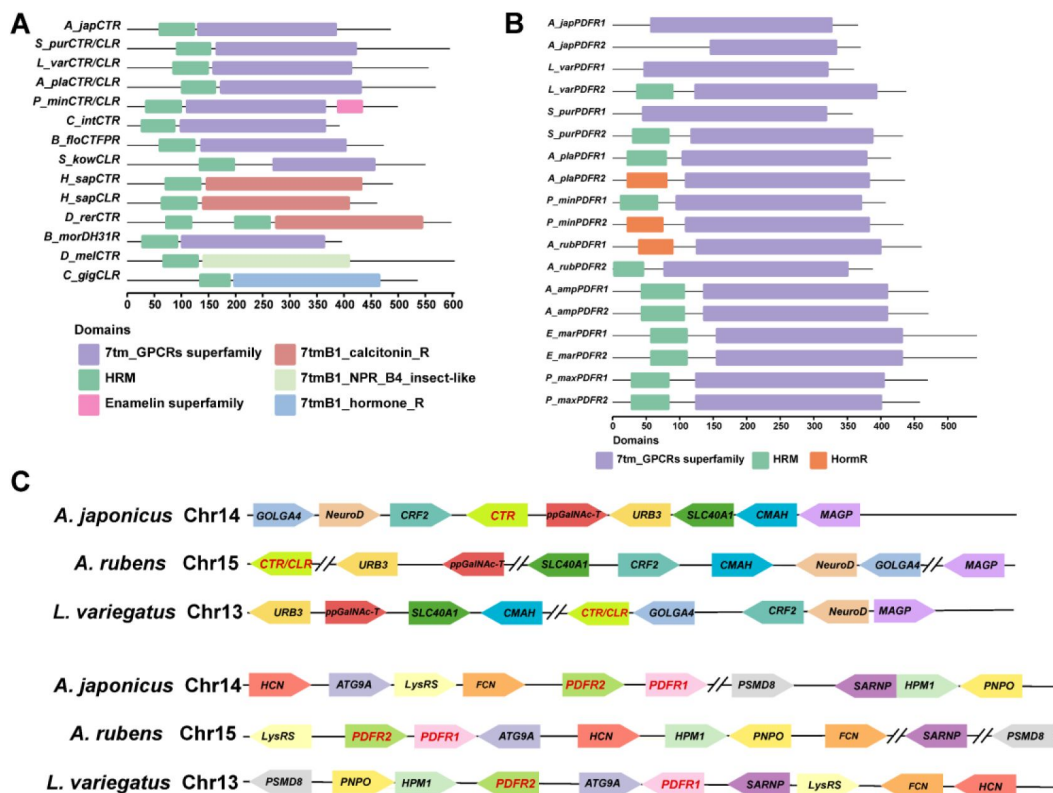


Figure 1-figure supplement 4.

Characterization of AjCT1 and AjCT2 receptors (AjCTR, AjPDR1 and AjPDR2). (A) Conserved domains of CTR/CLR-type protein sequences. Black line represents the length of proteins and different domains are represented by rectangles. (B) Conserved domains of PDRs in Bilateria. Black line represents the length of proteins and different domains are represented by rectangles. (C) Synteny analysis of AjCT1 and AjCT2 receptors (AjCTR, AjPDR1 and AjPDR2). The chromosomal locations of AjCT1 and AjCT2 receptor genes from *A. rubens*, *L. variegatus* and *A. japonicus* are shown, with different colors used to mark neighboring genes and with their transcriptional direction shown by arrows. *GOLGA4*, golgin subfamily A member 4; *NeuroD*, neurogenic differentiation factor; *CRF2*, corticotropin-releasing factor receptor 2; *CTR*, calcitonin type receptor; *CLR*, calcitonin gene-related peptide type 1 receptor; *ppGalNAc-T*, polypeptide N-acetylgalactosaminyltransferase; *URB3*, E3 ubiquitin-protein ligase ubr3; *SLC40A1*, solute carrier family 40 member 1; *CMAH*, cytidine monophosphate-N-acetylneuraminic acid hydroxylase; *MAGP*, microfibril-associated glycoprotein; *HCN*, hyperpolarization-gated and cyclic nucleotide regulated channel; *ATG9A*, autophagy-related protein 9A-like; *lysRS*, lysine--tRNA ligase; *FCN*, ficolin; *PDR1*, pigment-dispersing factor receptor 1; *PDR2*, pigment-dispersing factor receptor 2; *PSMD8*, 26S proteasome regulatory subunit 8; *SARNP*, SAP domain-containing ribonucleoprotein; *HPM1*, histidine protein methyltransferase 1; *PNPO*, pyridoxine-5'-phosphate oxidase.

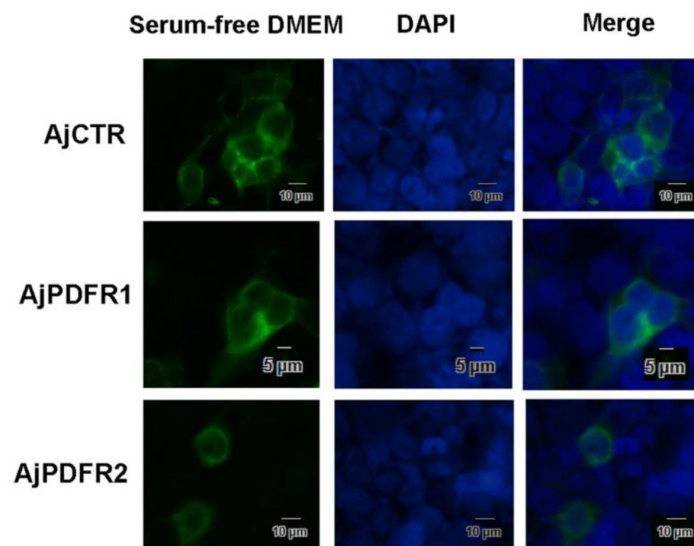
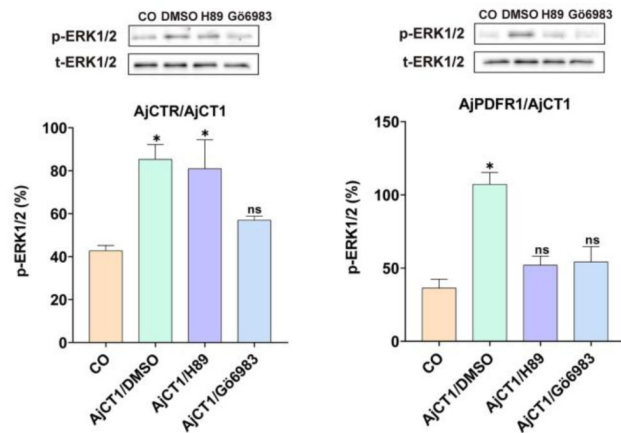


Figure 3-figure supplement 1.

Localization of AjCTR, AjPDFR1 and AjPDFR2 . Green fluorescence represents the localization of pEGFP-N1/receptors in cell membrane. Cell nucleus probe (DAPI) was applied for cell nucleus staining.

A



B

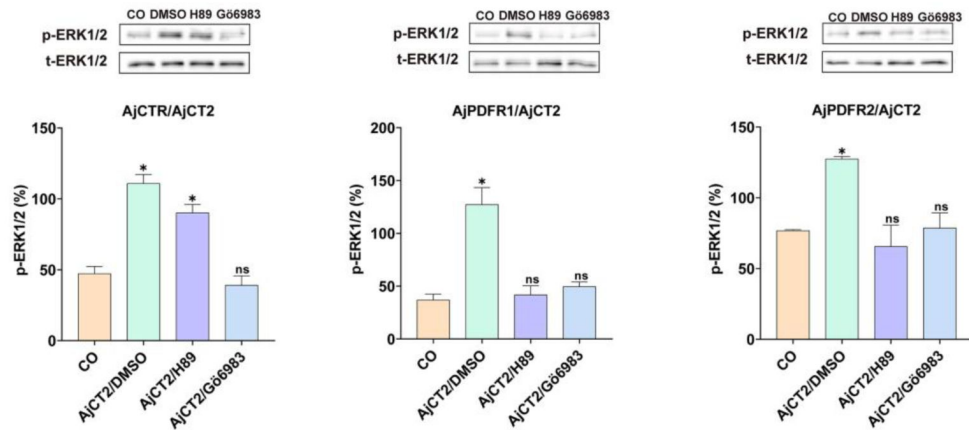


Figure 4-figure supplement 1.

Effects of PKA or PKC inhibitor on AjCT1 or AjCT2 stimulated ERK1/2 phosphorylation in AjCTR, AjPDFR1 or AjPDFR2 expressing HEK293T cells. (A) ERK1/2 phosphorylation, activated by AjCT1, is blocked by PKC inhibitor G66983 in AjCTR expressing HEK293T cells and is blocked by PKA inhibitor H89 and PKC inhibitor in AjPDFR1 expressing HEK293T cells. (B) ERK1/2 phosphorylation, activated by AjCT2, is blocked by PKC inhibitor in AjCTR expressing HEK293T cells and is blocked by PKA inhibitor and PKC inhibitor in AjPDFR1 and AjPDFR2 expressing HEK293T cells. Serum-starved HEK293 cells were pre-treated with DMSO, PKA inhibitor, or PKC inhibitor before AjCT1 or AjCT2 stimulation. All pictures and data are representative for at least three independent experiments.

Figure 5-figure supplement 1.

Negative control of circumoral nervous system *in situ* hybridization. The sense probe for AjCTP1/2 was incubated with sections of tissue from adult *A. japonicus* instead of an antisense probe. Scale bar: 100 μ m.

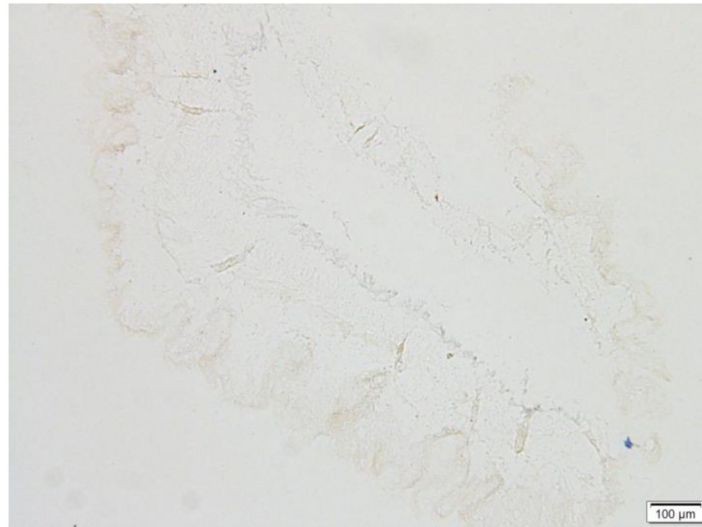
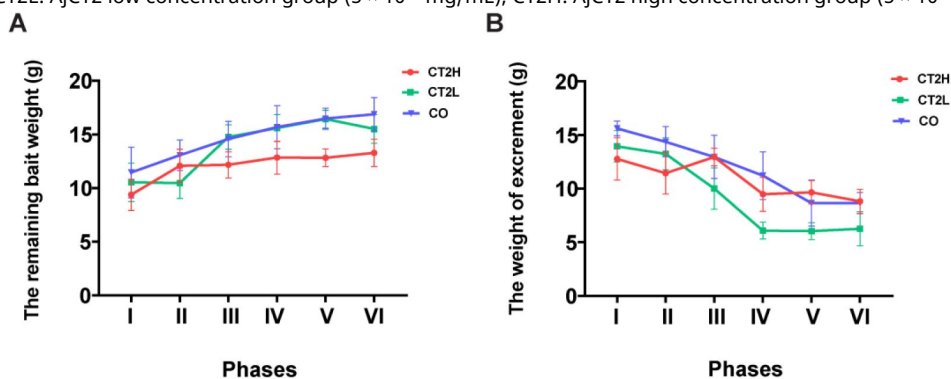


Figure 7A-figure supplement 1.

The mass of remaining bait and the excrement. (A) Calculation of the remaining bait mass in different phases. (B) The mass of excrement in different phases. Every four days was divided into one phase (I-VI) for statistics. CO: control group (sterilized seawater); CT2L: AjCT2 low concentration group (5×10^{-3} mg/mL); CT2H: AjCT2 high concentration group (5×10^{-1} mg/mL).



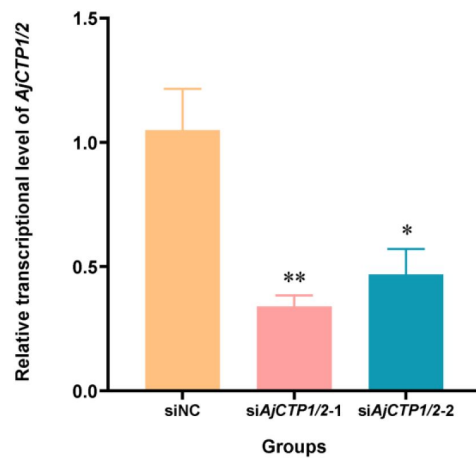


Figure 8-figure supplement 1.

The relative expression level of *AjCTP1/2* in siNC, siAjCTP1/2-1 and siAjCTP1/2-2. Values are means $\pm$ SEM ($n = 5$). Asterisks represent the significant difference as follows: *: $p < 0.05$, **: $p < 0.01$. siNC: negative control group; siAjCTP1/2-1 and siAjCTP1/2-2: experimental groups targeting *AjCTP1/2*.

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Editors

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Reviewer #1 (Public review):

Summary:

The manuscript characterizes a functional peptidergic system in the echinoderm *Apostichopus japonicus* that is related to the widely conserved family of calcitonin/diuretic hormone 31 (CT/DH31) peptides in bilaterian animals. In vitro analysis of receptor-ligand interactions, using multiple receptor activation assays, identifies three cognate receptors for two CT-like peptides in the sea cucumber, which stimulate cAMP, calcium, and ERK signaling. Only one of these receptors is closely related to the family of calcitonin and calcitonin-like receptors (CTR/CLR) in bilaterian animals, whereas two other receptors cluster with invertebrate pigment dispersing factor receptors (PDFRs). In addition, this study sheds light on the transcript expression and in vivo functions of CT-like peptides in *A. japonicus*, by quantitative real-time PCR, in situ hybridization, pharmacological experiments on body wall muscle and intestine preparations, and peptide injection and RNAi knockdown experiments. This reveals a conserved function of CT-like peptides as muscle relaxants and hints at a potential role as growth regulators in *A. japonicus*.

Strengths:

This work combines both in vitro and in vivo functional assays to identify a CT-like peptidergic system in an economically relevant echinoderm species, the sea cucumber *A. japonicus*. A major strength of the study is that it identifies three G protein-coupled receptors for AjCT-like peptides, one related to the CTR/CLR family and two related to the PDFR family. A similar finding was previously reported for the CT-related peptide DH31 in *Drosophila melanogaster* that activates both CT-type and PDF-type receptors. Here, the authors expand this observation to a deuterostomian animal, which suggests that receptor promiscuity is a more general feature of the CT/DH31 peptide family and that CT/DH31-like peptides may activate both CT-type and PDF-type receptors in other animals as well.

Besides the identification of receptor-ligand pairs, the downstream signaling pathways of AjCT receptors have been characterized, highlighting broad effects on cAMP, calcium, and ERK signaling. Functional characterization of the CT-related peptide system in heterologous cells is complemented with ex vivo and in vivo experiments. First, peptide injection and RNAi knockdown experiments establish transcriptional regulation of all three identified receptors in response to changing AjCT peptide levels. Second, ex vivo experiments reveal a conserved role for the two CT-like peptides as muscle relaxants, which have differential effects on body wall muscle and intestine preparations. Finally, peptide injection studies suggest a putative role for one of the two CT-like peptides (AjCT2) in growth regulation.

Weaknesses:

Analysis of transcript expression is limited to the CT-peptide encoding gene, while no gene expression analysis was attempted for the three identified receptors. Differences in the activation of downstream signaling pathways between the three receptors are also

questionable due to unclarities in the statistical analysis and variation in the control and experimental data in heterologous assays. Together, this makes it difficult to propose a mechanism underlying differences in the functions of the two CT-like peptides in muscle control and growth regulation.

The authors also suggest a putative orexigenic role for the CT-like peptidergic system in feeding behavior. This effect is not well supported by the experimental data provided, as no detailed analysis of feeding behavior was carried out (only indirect measurements were performed that could be influenced by other peptidergic effects, such as on muscle relaxation) and no statistically significant differences were reported in these assays.

Overall, details regarding statistical analyses are not (clearly) specified in the manuscript, and there are several instances where statements are not supported by literature evidence.

<https://doi.org/10.7554/eLife.101799.1.sa2>

Reviewer #2 (Public review):

Summary:

The authors show that *A. japonicus* calcitonins (AjCT1 and AjCT2) activate not only the calcitonin/calcitonin-like receptor but also activate the two PDF receptors, *ex vivo*. They also explore secondary messenger pathways that are recruited following receptor activation. They determine the source of CT1 and CT2 using qPCR and *in situ* hybridization and finally test the effects of these peptides on tissue contractions, feeding, and growth. This study provides solid evidence that CT1 and CT2 act as ligands for calcitonin receptors; however, evidence supporting cross-talk between CT peptides and PDF receptors is only based on *ex vivo* experiments.

Strengths:

This is the first study to report the pharmacological characterization of CT receptors in an echinoderm. Multiple lines of evidence in cell culture (receptor internalization and secondary messenger pathways) support this conclusion.

Weaknesses:

The authors claim that *A. japonicus* CTs activate "PDF" receptors and suggest that this cross-talk is evolutionarily ancient since a similar phenomenon also exists in the fly *Drosophila melanogaster*. These conclusions are not fully supported for several reasons. The authors perform phylogenetic analysis to show that the two "PDF" receptors form an independent clade. This clade is sister to the clade comprising CT receptors. This phylogenetic analysis suffers from several issues. Firstly, the phylogenies lack bootstrap support. Secondly, the resolution of the phylogeny is poor because representative members from diverse phyla have not been included. For instance, insect or other protostomian PDF receptors have not been included so how can the authors distinguish between "PDF" receptors or another group of CT receptors? Thirdly, no *in vivo* evidence has been presented to support that CT can activate "PDF" receptors *in vivo*.

The source of CT which mediates the effects on longitudinal muscles and intestine is unclear. Is it autocrine or paracrine signaling by CT from the same tissue or is it long-range hormonal signaling?

Pharmacology experiments showing the effects of CT1 and CT2 on ACh-induced contractions were performed. Sample traces have been provided but no traces with ACh alone have been included. How long do ACh-induced contractions persist? These controls are necessary to

differentiate between the eventual decay of ACh effects and relaxation induced by CT1 and CT2. The traces also do not reflect the results portrayed in dose-response curves. For instance, in Figure 6B, maximum relaxation is reported for 10-6M. Yet, the trace hardly shows any difference before and after the addition of 10-6M peptide. The maximum effect in the trace appears to be after the addition of 10-8M peptide.

I am unsure how differences in wet mass indicate feeding and growth differences since no justification has been provided. Couldn't wet mass also be influenced by differences in osmotic balance, a key function of calcitonin-like peptides in protostomian invertebrates? The statistical comparisons have not been included in Figure 7B.

While the authors succeeded in knocking down CT, the physiological effects of reduced CT signaling were not examined.

<https://doi.org/10.7554/eLife.101799.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

*The manuscript characterizes a functional peptidergic system in the echinoderm *Apostichopus japonicus* that is related to the widely conserved family of calcitonin/diuretic hormone 31 (CT/DH31) peptides in bilaterian animals. In vitro analysis of receptor-ligand interactions, using multiple receptor activation assays, identifies three cognate receptors for two CT-like peptides in the sea cucumber, which stimulate cAMP, calcium, and ERK signaling. Only one of these receptors is closely related to the family of calcitonin and calcitonin-like receptors (CTR/CLR) in bilaterian animals, whereas two other receptors cluster with invertebrate pigment dispersing factor receptors (PDFRs). In addition, this study sheds light on the transcript expression and in vivo functions of CT-like peptides in *A. japonicus*, by quantitative real-time PCR, in situ hybridization, pharmacological experiments on body wall muscle and intestine preparations, and peptide injection and RNAi knockdown experiments. This reveals a conserved function of CT-like peptides as muscle relaxants and hints at a potential role as growth regulators in *A. japonicus*.*

Strengths:

*This work combines both in vitro and in vivo functional assays to identify a CT-like peptidergic system in an economically relevant echinoderm species, the sea cucumber *A. japonicus*. A major strength of the study is that it identifies three G protein-coupled receptors for AjCT-like peptides, one related to the CTR/CLR family and two related to the PDFR family. A similar finding was previously reported for the CT-related peptide DH31 in *Drosophila melanogaster* that activates both CT-type and PDF-type receptors. Here, the authors expand this observation to a deuterostomian animal, which suggests that receptor promiscuity is a more general feature of the CT/DH31 peptide family and that CT/DH31-like peptides may activate both CT-type and PDF-type receptors in other animals as well.*

Besides the identification of receptor-ligand pairs, the downstream signaling pathways of AjCT receptors have been characterized, highlighting broad effects on cAMP, calcium, and ERK signaling. Functional characterization of the CT-related peptide system in heterologous cells is complemented with ex vivo and in vivo experiments. First, peptide

injection and RNAi knockdown experiments establish transcriptional regulation of all three identified receptors in response to changing AjCT peptide levels. Second, ex vivo experiments reveal a conserved role for the two CT-like peptides as muscle relaxants, which have differential effects on body wall muscle and intestine preparations. Finally, peptide injection studies suggest a putative role for one of the two CT-like peptides (AjCT2) in growth regulation.

Weaknesses:

(1) Analysis of transcript expression is limited to the CT-peptide encoding gene, while no gene expression analysis was attempted for the three identified receptors. Differences in the activation of downstream signaling pathways between the three receptors are also questionable due to unclarities in the statistical analysis and variation in the control and experimental data in heterologous assays. Together, this makes it difficult to propose a mechanism underlying differences in the functions of the two CT-like peptides in muscle control and growth regulation.

Thank you for the reviewer's comment, we will supplement the expression analysis for the three identified receptors. Actually, we did all the statistical tests for all the experiments, and maybe the form of marking is a bit messy, so sorry for the confusion and we will uniform them and include all this information both in the figure legends and the methods section. And for the variation in the control and experimental data, because every control is transfected with different receptors or uses cells from different batches, that is why the control conditions in different experiments have little bit variation.

(2) The authors also suggest a putative orexigenic role for the CT-like peptidergic system in feeding behavior. This effect is not well supported by the experimental data provided, as no detailed analysis of feeding behavior was carried out (only indirect measurements were performed that could be influenced by other peptidergic effects, such as on muscle relaxation) and no statistically significant differences were reported in these assays.

Thank you for the reviewer's comment. Actually, we did all the statistical tests for all the experiments, the mass of remaining bait and the excrement were added in Figure 7A-figure supplement 1 and we will conduct additional behavioral experiments to explore the changes in feeding behavior of *A. japonicus* after injection of CT-type neuropeptides to support the role of CT-like peptidergic system in the regulation of feeding behavior, as I mentioned above, maybe the form of marking is a bit messy, so sorry for the confusion and we will uniform them and include all this information both in the figure legends and the methods section. And also we will supplement the experiments to further support our claim by assessing the feeding and growth factors after knocking down the CTP encoding genes.

(3) Overall, details regarding statistical analyses are not (clearly) specified in the manuscript, and there are several instances where statements are not supported by literature evidence.

Again, actually, we did all the statistical tests for all the experiments, as I mentioned above, maybe the form of marking is a bit messy, so sorry for the confusion and we will uniform them and include all this information both in the figure legends and the methods section. And we will also supplement more experiments and add more literature evidence to support our statements.

Reviewer #2 (Public review):

Summary:

*The authors show that *A. japonicus* calcitonins (AjCT1 and AjCT2) activate not only the calcitonin/calcitonin-like receptor but also activate the two PDF receptors, ex vivo. They also explore secondary messenger pathways that are recruited following receptor activation. They determine the source of CT1 and CT2 using qPCR and in situ hybridization and finally test the effects of these peptides on tissue contractions, feeding, and growth. This study provides solid evidence that CT1 and CT2 act as ligands for calcitonin receptors; however, evidence supporting cross-talk between CT peptides and PDF receptors is only based on ex vivo experiments.*

Strengths:

This is the first study to report the pharmacological characterization of CT receptors in an echinoderm. Multiple lines of evidence in cell culture (receptor internalization and secondary messenger pathways) support this conclusion.

Weaknesses:

*(1) The authors claim that *A. japonicus* CTs activate "PDF" receptors and suggest that this cross-talk is evolutionarily ancient since a similar phenomenon also exists in the fly *Drosophila melanogaster*. These conclusions are not fully supported for several reasons. The authors perform phylogenetic analysis to show that the two "PDF" receptors form an independent clade. This clade is sister to the clade comprising CT receptors. This phylogenetic analysis suffers from several issues. Firstly, the phylogenies lack bootstrap support. Secondly, the resolution of the phylogeny is poor because representative members from diverse phyla have not been included. For instance, insect or other protostomian PDF receptors have not been included so how can the authors distinguish between "PDF" receptors or another group of CT receptors? Thirdly, no in vivo evidence has been presented to support that CT can activate "PDF" receptors in vivo.*

We do agree with the reviewer that the cross-talk between CTs and PDFRs is not so solid based on our current study.

So firstly, we will re-do the phylogenetic analyses as the reviewer suggested and mark the bootstrap value, then we will supplement more experiments (like PDFR knockdown) to further confirm that CT can activate "PDF" receptors in vivo.

(2) The source of CT which mediates the effects on longitudinal muscles and intestine is unclear. Is it autocrine or paracrine signaling by CT from the same tissue or is it long-range hormonal signaling?

Thank you for the reviewer's comment, actually we have done in situ and immunohistochemical experiments for CTP and CT in different tissues, we just did not put them in our current manuscript, we will add them in the revised version.

(3) Pharmacology experiments showing the effects of CT1 and CT2 on ACh-induced contractions were performed. Sample traces have been provided but no traces with ACh alone have been included. How long do ACh-induced contractions persist? These controls are necessary to differentiate between the eventual decay of ACh effects and relaxation induced by CT1 and CT2. The traces also do not reflect the results portrayed in dose-response curves. For instance, in Figure 6B, maximum relaxation is reported for 10-6M. Yet, the trace hardly shows any difference before and after the addition of 10-6M peptide. The maximum effect in the trace appears to be after the addition of 10-8M peptide.

Thank you for the reviewer's comment, we will provide the trace of contraction caused by ACh alone. In Figure 6B, the trace represents successive treatments of neuropeptides at different concentrations, which represents a cumulative effect. Therefore, the corresponding receptors may become desensitized when high concentration of peptide is finally applied. Actually, we examined the pharmacological effects of CT2 at 10^{-6} M concentrations, which exhibited the maximum relaxation, and we will provide this trace.

(4) I am unsure how differences in wet mass indicate feeding and growth differences since no justification has been provided. Couldn't wet mass also be influenced by differences in osmotic balance, a key function of calcitonin-like peptides in protostomian invertebrates? The statistical comparisons have not been included in Figure 7B.

Thank you for the reviewer's comment, we will analyze the weight gain rate, growth rate and feeding rate of *A. japonicus* to explain the difference of feeding and growth between injection group and control group. And we can confirm that wet mass is not influenced by differences in osmotic balance, we will put our supporting evidence in supplementary files in revised manuscript and we did not find the key function of calcitonin-like peptides observed in protostomian invertebrates. And we will include the statistical comparisons in Figure 7B.

(5) While the authors succeeded in knocking down CT, the physiological effects of reduced CT signaling were not examined.

Thank you for the reviewer's insightful suggestion, we will supplement the experiments about the physiological effects after knocking down CT.

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